

Pathway Analysis Report

This report contains the pathway analysis results for the submitted sample ". Analysis was performed against Reactome version 95 on 19/12/2025. The web link to these results is:

<https://reactome.org/PathwayBrowser/#/ANALYSIS=MjAyNTEyMTkwODU4NDhfNTM0Ng%3D%3D>

Please keep in mind that analysis results are temporarily stored on our server. The storage period depends on usage of the service but is at least 7 days. As a result, please note that this URL is only valid for a limited time period and it might have expired.

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1. Introduction

Reactome is a curated database of pathways and reactions in human biology. Reactions can be considered as pathway 'steps'. Reactome defines a 'reaction' as any event in biology that changes the state of a biological molecule. Binding, activation, translocation, degradation and classical biochemical events involving a catalyst are all reactions. Information in the database is authored by expert biologists, entered and maintained by Reactome's team of curators and editorial staff. Reactome content frequently cross-references other resources e.g. NCBI, Ensembl, UniProt, KEGG (Gene and Compound), ChEBI, PubMed and GO. Orthologous reactions inferred from annotation for Homo sapiens are available for 14 non-human species including mouse, rat, chicken, puffer fish, worm, fly and yeast. Pathways are represented by simple diagrams following an SBGN-like format.

Reactome's annotated data describe reactions possible if all annotated proteins and small molecules were present and active simultaneously in a cell. By overlaying an experimental dataset on these annotations, a user can perform a pathway over-representation analysis. By overlaying quantitative expression data or time series, a user can visualize the extent of change in affected pathways and its progression. A binomial test is used to calculate the probability shown for each result, and the p-values are corrected for the multiple testing (Benjamini-Hochberg procedure) that arises from evaluating the submitted list of identifiers against every pathway.

To learn more about our Pathway Analysis, please have a look at our relevant publications:

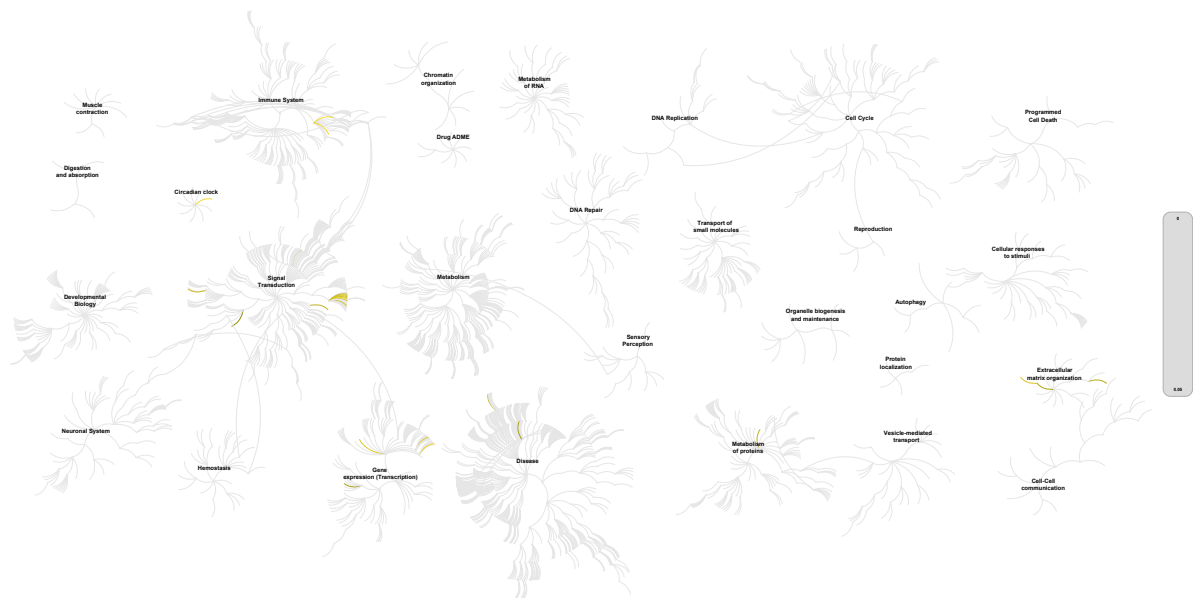
Fabregat A, Sidiropoulos K, Garapati P, Gillespie M, Hausmann K, Haw R, ... D'Eustachio P (2016). The reactome pathway knowledgebase. *Nucleic Acids Research*, 44(D1), D481–D487. <https://doi.org/10.1093/nar/gkv1351>. 

Fabregat A, Sidiropoulos K, Viteri G, Forner O, Marin-Garcia P, Arnau V, ... Hermjakob H (2017). Reactome pathway analysis: a high-performance in-memory approach. *BMC Bioinformatics*, 18. 

2. Properties

- This is an **overrepresentation** analysis: A statistical (hypergeometric distribution) test that determines whether certain Reactome pathways are over-represented (enriched) in the submitted data. It answers the question 'Does my list contain more proteins for pathway X than would be expected by chance?' This test produces a probability score, which is corrected for false discovery rate using the Benjamini-Hochberg method. [↗](#)
- 472 out of 576 identifiers in the sample were found in Reactome, where 2012 pathways were hit by at least one of them.
- All non-human identifiers have been converted to their human equivalent. [↗](#)
- IntAct interactors were included to increase the analysis background. This greatly increases the size of Reactome pathways, which maximises the chances of matching your submitted identifiers to the expanded pathway, but will include interactors that have not undergone manual curation by Reactome and may include interactors that have no biological significance, or unexplained relevance.
- This report is filtered to show only results for species 'Homo sapiens' and resource 'all resources'.
- The unique ID for this analysis (token) is MjAyNTEyMTkwODU4NDhfNTM0Ng%3D%3D. This ID is valid for at least 7 days in Reactome's server. Use it to access Reactome services with your data.

3. Genome-wide overview



This figure shows a genome-wide overview of the results of your pathway analysis. Reactome pathways are arranged in a hierarchy. The center of each of the circular "bursts" is the root of one top-level pathway, for example "DNA Repair". Each step away from the center represents the next level lower in the pathway hierarchy. The color code denotes over-representation of that pathway in your input dataset. Light grey signifies pathways which are not significantly over-represented.

4. Most significant pathways

The following table shows the 25 most relevant pathways sorted by p-value.

Pathway name	Entities				Reactions	
	found	ratio	p-value	FDR*	found	ratio
Interleukin-10 signaling	24 / 181	0.007	6.76e-04	0.993	6 / 15	9.45e-04
The CRY:PER:kinase complex represses transactivation by the BMAL:CLOCK (ARNTL:CLOCK) complex	5 / 21	8.46e-04	0.003	0.993	4 / 4	2.52e-04
Interleukin-4 and Interleukin-13 signaling	45 / 473	0.019	0.005	0.993	15 / 47	0.003
TFAP2 (AP-2) family regulates transcription of other transcription factors	3 / 8	3.22e-04	0.006	0.993	2 / 2	1.26e-04
RHOB GTPase cycle	9 / 75	0.003	0.007	0.993	3 / 6	3.78e-04
Nuclear Receptor transcription pathway	12 / 126	0.005	0.012	0.993	2 / 2	1.26e-04
RUNX3 regulates RUNX1-mediated transcription	2 / 4	1.61e-04	0.014	0.993	2 / 2	1.26e-04
TGFBR1 LBD Mutants in Cancer	2 / 4	1.61e-04	0.014	0.993	1 / 1	6.30e-05
Molecules associated with elastic fibres	6 / 44	0.002	0.015	0.993	8 / 10	6.30e-04
RAC3 GTPase cycle	10 / 100	0.004	0.016	0.993	3 / 6	3.78e-04
RAC2 GTPase cycle	11 / 101	0.004	0.017	0.993	5 / 10	6.30e-04
RHOC GTPase cycle	10 / 87	0.004	0.018	0.993	4 / 6	3.78e-04
PI3K events in ERBB2 signaling	4 / 22	8.86e-04	0.018	0.993	6 / 7	4.41e-04
RHOJ GTPase cycle	7 / 59	0.002	0.018	0.993	2 / 3	1.89e-04
Signaling by FGFR1 amplification mutants	2 / 5	2.01e-04	0.021	0.993	5 / 5	3.15e-04
Adenosine P1 receptors	2 / 5	2.01e-04	0.021	0.993	2 / 2	1.26e-04
RHOA GTPase cycle	13 / 154	0.006	0.023	0.993	4 / 6	3.78e-04
Antagonism of Activin by Follistatin	3 / 14	5.64e-04	0.025	0.993	2 / 2	1.26e-04
Post-translational protein phosphorylation	10 / 109	0.004	0.027	0.993	1 / 1	6.30e-05
RAC1 GTPase cycle	15 / 191	0.008	0.027	0.993	3 / 6	3.78e-04
Anchoring fibril formation	3 / 15	6.04e-04	0.03	0.993	4 / 4	2.52e-04
Elastic fibre formation	8 / 83	0.003	0.035	0.993	15 / 17	0.001
Post-transcriptional silencing by small RNAs	2 / 7	2.82e-04	0.04	0.993	3 / 3	1.89e-04
Loss of Function of TGFBR1 in Cancer	2 / 7	2.82e-04	0.04	0.993	2 / 2	1.26e-04

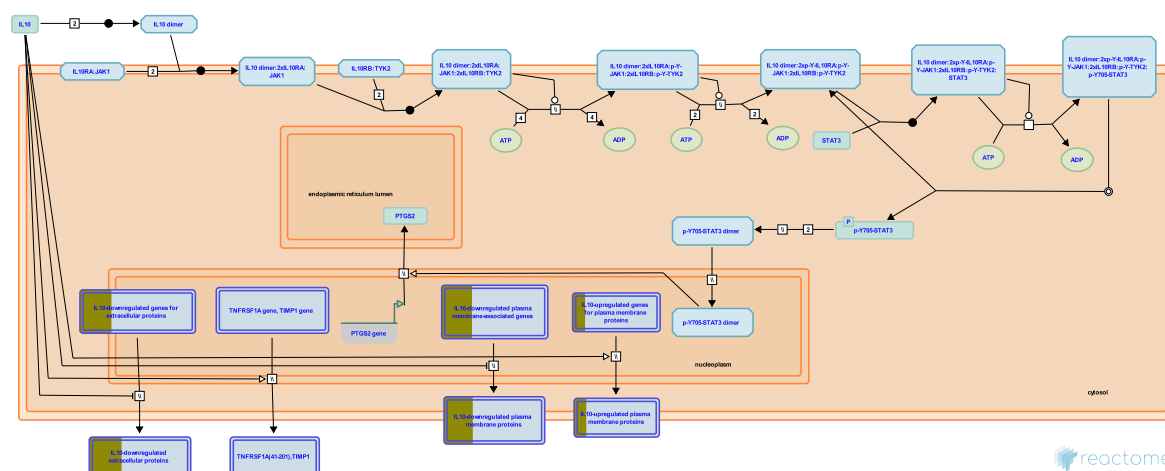
Pathway name	Entities				Reactions	
	found	ratio	p-value	FDR*	found	ratio
G alpha (12/13) signalling events	11 / 121	0.005	0.048	0.993	9 / 15	9.45e-04

* False Discovery Rate

5. Pathways details

For every pathway of the most significant pathways, we present its diagram, as well as a short summary, its bibliography and the list of inputs found in it.

1. Interleukin-10 signaling (R-HSA-6783783)



Interleukin-10 (IL10) was originally described as a factor named cytokine synthesis inhibitory factor that inhibited T-helper (Th) 1 activation and Th1 cytokine production (Fiorentino et al. 1989). It was found to be expressed by a variety of cell types including macrophages, dendritic cell subsets, B cells, several T-cell subpopulations including Th2 and T-regulatory cells (Tregs) and Natural Killer (NK) cells (Moore et al. 2001). It is now recognized that the biological effects of IL10 are directed at antigen-presenting cells (APCs) such as macrophages and dendritic cells (DCs), its effects on T-cell development and differentiation are largely indirect via inhibition of macrophage/dendritic cell activation and maturation (Pestka et al. 2004, Mocellin et al. 2004). T cells are thought to be the main source of IL10 (Hedrich & Bream 2010). IL10 inhibits a broad spectrum of activated macrophage/monocyte functions including monokine synthesis, NO production, and expression of class II MHC and costimulatory molecules such as IL12 and CD80/CD86 (de Waal Malefyt et al. 1991, Gazzinelli et al. 1992). Studies with recombinant cytokine and neutralizing antibodies revealed pleiotropic activities of IL10 on B, T, and mast cells (de Waal Malefyt et al. 1993, Rousset et al. 1992, Thompson-Snipes et al. 1991) and provided evidence for the *in vivo* significance of IL10 activities (Ishida et al. 1992, 1993). IL10 antagonizes the expression of MHC class II and the co-stimulatory molecules CD80/CD86 as well as the pro-inflammatory cytokines IL1Beta, IL6, IL8, TNFalpha and especially IL12 (Fiorentino et al. 1991, D'Andrea et al. 1993). The biological role of IL10 is not limited to inactivation of APCs, it also enhances B cell, granulocyte, mast cell, and keratinocyte growth/differentiation, as well as NK-cell and CD8+ cytotoxic T-cell activation (Moore et al. 2001, Hedrich & Bream 2010). IL10 also enhances NK-cell proliferation and/or production of IFN-gamma (Cai et al. 1999).

IL10-deficient mice exhibited inflammatory bowel disease (IBD) and other exaggerated inflammatory responses (Kuhn et al. 1993, Berg et al. 1995) indicating a critical role for IL10 in limiting inflammatory responses. Dysregulation of IL10 is linked with susceptibility to numerous infectious and autoimmune diseases in humans and mouse models (Hedrich & Bream 2010).

IL10 signaling is initiated by binding of homodimeric IL10 to the extracellular domains of two ad-joining IL10RA molecules. This tetramer then binds two IL10RB chains. IL10RB cannot bind to IL10 unless bound to IL10RA (Ding et al. 2001, Yoon et al. 2006); binding of IL10 to IL10RA without the co-presence of IL10RB fails to initiate signal transduction (Kotenko et al. 1997).

IL10 binding activates the receptor-associated Janus tyrosine kinases, JAK1 and TYK2, which are constitutively bound to IL10R1 and IL10R2 respectively. In the classic model of receptor activation assembly of the receptor complex is believed to enable JAK1/TYK2 to phosphorylate and activate each other. Alternatively the binding of IL10 may cause conformational changes that allow the pseudokinase inhibitory domain of one JAK kinase to move away from the kinase domain of the other JAK within the receptor dimer-JAK complex, allowing the two kinase domains to interact and trans-activate (Waters & Brooks 2015).

The activated JAK kinases phosphorylate the intracellular domains of the IL10R1 chains on specific tyrosine residues. These phosphorylated tyrosine residues and their flanking peptide sequences serve as temporary docking sites for the latent, cytosolic, transcription factor, STAT3. STAT3 transiently docks on the IL10R1 chain via its SH2 domain, and is in turn tyrosine phosphorylated by the receptor-associated JAKs. Once activated, it dissociates from the receptor, dimerizes with other STAT3 molecules, and translocates to the nucleus where it binds with high affinity to STAT-binding elements (SBEs) in the promoters of IL-10-inducible genes (Donnelly et al. 1999).

References

Moore KW, de Waal Malefyt R, Coffman RL & O'Garra A (2001). Interleukin-10 and the interleukin-10 receptor. *Annu. Rev. Immunol.*, 19, 683-765. [↗](#)

Edit history

Date	Action	Author
2015-06-17	Authored	Jupe S
2015-06-17	Created	Jupe S
2016-09-05	Reviewed	Meldal BH
2016-11-14	Edited	Jupe S

10 submitted entities found in this pathway, mapping to 20 Reactome entities

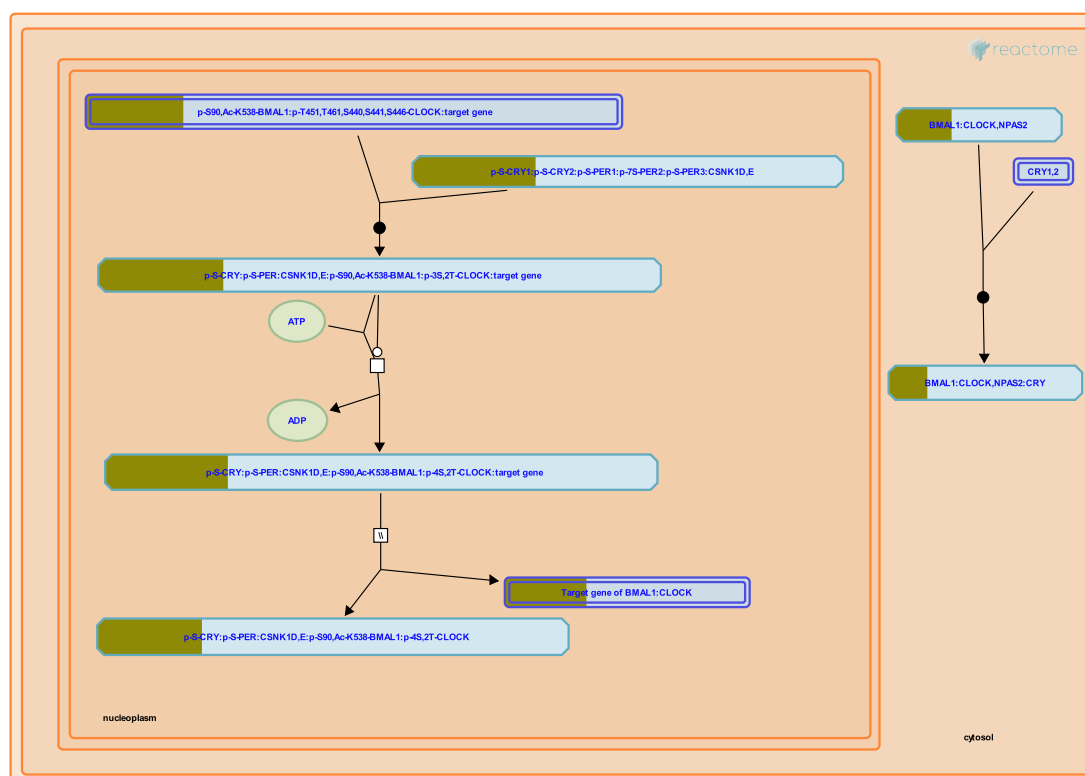
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CSF1	P09603	CXCL1	P09341	CXCL2	P19875
CXCL3	P19875	CXCL8	P10145	FPR1	P21462
ICAM1	P05362	IL1B	P01583, P01584	IL6	P05231
LIF	P15018				

Input	Ensembl Id	Input	Ensembl Id	Input	Ensembl Id
CSF1	ENSG00000184371	CXCL1	ENSG00000163739	CXCL2	ENSG00000081041
CXCL8	ENSG00000169429	FPR1	ENSG00000171051	ICAM1	ENSG00000090339
IL1B	ENSG00000125538	IL6	ENSG00000136244	LIF	ENSG00000128342

Interactors found in this pathway (4)

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
MYC	EBI-1265559	P40763	NFIX	Q14938-4, Q14938-3	P40763
PIK3R1	P27986-2	P40763	PTK2B	Q14289	P40763

2. The CRY:PER:kinase complex represses transactivation by the BMAL:CLOCK (ARNTL:CLOCK) complex (R-HSA-9931521)



A complex containing Cryptochrome proteins CRY1, CRY2, Period proteins PER1, PER2, PER3, and kinases CSNK1D and/or CSNK1E binds a heterodimer containing phosphorylated CLOCK and phosphorylated, acetylated BMAL1 bound to E-boxes in the promoters of target genes (inferred from mouse homologs in Griffin et al. 1999, Langmesser et al. 2008, Ye et al. 2011, Ye et al. 2014, Aryal et al. 2017, Cao et al. 2021). This effectively delivers the kinases CSNK1D and/or CSNK1E to the proximity of CLOCK, resulting in further phosphorylation of CLOCK (inferred from mouse homologs in Cao et al. 2021). NPAS2, which acts redundantly with CLOCK, may be similarly phosphorylated. The phosphorylation of CLOCK causes the dissociation of the BMAL1:CLOCK heterodimer from DNA (inferred from mouse homologs in Ye et al. 2014, Chiou et al. 2016, Cao et al. 2021), terminating the diurnal activation of gene expression by BMAL1:CLOCK.

References

- Ye R, Selby CP, Ozturk N, Annayev Y & Sancar A (2011). Biochemical analysis of the canonical model for the mammalian circadian clock. *J Biol Chem*, 286, 25891-902. [🔗](#)
- Aryal RP, Kwak PB, Tamayo AG, Gebert M, Chiu PL, Walz T & Weitz CJ (2017). Macromolecular Assemblies of the Mammalian Circadian Clock. *Mol Cell*, 67, 770-782.e6. [🔗](#)
- Ye R, Selby CP, Chiou YY, Ozkan-Dagliyan I, Gaddameedhi S & Sancar A (2014). Dual modes of CLOCK:BMAL1 inhibition mediated by Cryptochrome and Period proteins in the mammalian circadian clock. *Genes Dev*, 28, 1989-98. [🔗](#)
- Cao X, Yang Y, Selby CP, Liu Z & Sancar A (2021). Molecular mechanism of the repressive phase of the mammalian circadian clock. *Proc Natl Acad Sci U S A*, 118. [🔗](#)

Langmesser S, Tallone T, Bordon A, Rusconi S & Albrecht U (2008). Interaction of circadian clock proteins PER2 and CRY with BMAL1 and CLOCK. BMC Mol Biol, 9, 41. [🔗](#)

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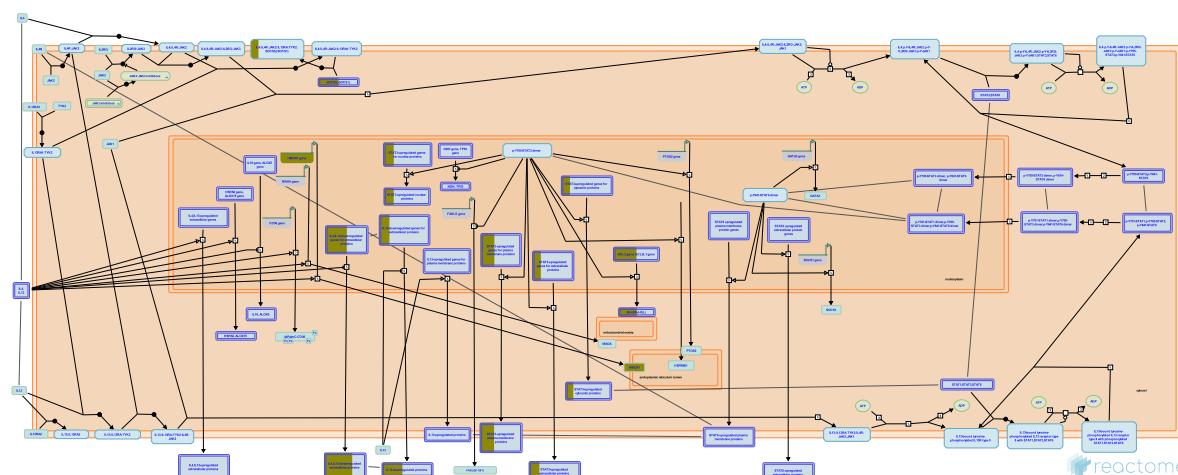
Date	Action	Author
2024-12-09	Edited	May B
2024-12-09	Authored	May B
2024-12-12	Created	May B
2025-03-04	Reviewed	Sancar A, Albrecht U

3 submitted entities found in this pathway, mapping to 5 Reactome entities

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
NPAS2	Q99743	PER1	O15534	PER3	P56645

Input	Ensembl Id	Input	Ensembl Id
PER1	ENSG00000179094	PER3	ENSG00000049246

3. Interleukin-4 and Interleukin-13 signaling (R-HSA-6785807)



Interleukin-4 (IL4) is a principal regulatory cytokine during the immune response, crucially important in allergy and asthma (Nelms et al. 1999). When resting T cells are antigen-activated and expand in response to Interleukin-2 (IL2), they can differentiate as Type 1 (Th1) or Type 2 (Th2) T helper cells. The outcome is influenced by IL4. Th2 cells secrete IL4, which both stimulates Th2 in an autocrine fashion and acts as a potent B cell growth factor to promote humoral immunity (Nelms et al. 1999).

Interleukin-13 (IL13) is an immunoregulatory cytokine secreted predominantly by activated Th2 cells. It is a key mediator in the pathogenesis of allergic inflammation. IL13 shares many functional properties with IL4, stemming from the fact that they share a common receptor subunit. IL13 receptors are expressed on human B cells, basophils, eosinophils, mast cells, endothelial cells, fibroblasts, monocytes, macrophages, respiratory epithelial cells, and smooth muscle cells, but unlike IL4, not T cells. Thus IL13 does not appear to be important in the initial differentiation of CD4 T cells into Th2 cells, rather it is important in the effector phase of allergic inflammation (Hershey et al. 2003).

IL4 and IL13 induce “alternative activation” of macrophages, inducing an anti-inflammatory phenotype by signaling through IL4R alpha in a STAT6 dependent manner. This signaling plays an important role in the Th2 response, mediating anti-parasitic effects and aiding wound healing (Gordon & Martinez 2010, Loke et al. 2002)

There are two types of IL4 receptor complex (Andrews et al. 2006). Type I IL4R (IL4R1) is predominantly expressed on the surface of hematopoietic cells and consists of IL4R and IL2RG, the common gamma chain. Type II IL4R (IL4R2) is predominantly expressed on the surface of nonhematopoietic cells, it consists of IL4R and IL13RA1 and is also the type II receptor for IL13. (Obiri et al. 1995, Aman et al. 1996, Hilton et al. 1996, Miloux et al. 1997, Zhang et al. 1997). The second receptor for IL13 consists of IL4R and Interleukin-13 receptor alpha 2 (IL13RA2), sometimes called Interleukin-13 binding protein (IL13BP). It has a high affinity receptor for IL13 (Kd = 250 pmol/L) but is not sufficient to render cells responsive to IL13, even in the presence of IL4R (Donaldson et al. 1998). It is reported to exist in soluble form (Zhang et al. 1997) and when overexpressed reduces JAK-STAT signaling (Kawakami et al. 2001). It's function may be to prevent IL13 signalling via the functional IL4R:IL13RA1 receptor. IL13RA2 is overexpressed and enhances cell invasion in some human cancers (Joshi & Puri 2012).

The first step in the formation of IL4R1 (IL4:IL4R:IL2RB) is the binding of IL4 with IL4R (Hoffman et al. 1995, Shen et al. 1996, Hage et al. 1999). This is also the first step in formation of IL4R2 (IL4:IL4R:IL13RA1). After the initial binding of IL4 and IL4R, IL2RB binds (LaPorte et al. 2008), to form IL4R1. Alternatively, IL13RA1 binds, forming IL4R2. In contrast, the type II IL13 complex (IL13R2) forms with IL13 first binding to IL13RA1 followed by recruitment of IL4R (Wang et al. 2009).

Crystal structures of the IL4:IL4R:IL2RG, IL4:IL4R:IL13RA1 and IL13:IL4R:IL13RA1 complexes have been determined (LaPorte et al. 2008). Consistent with these structures, in monocytes IL4R is tyrosine phosphorylated in response to both IL4 and IL13 (Roy et al. 2002, Gordon & Martinez 2010) while IL13RA1 phosphorylation is induced only by IL13 (Roy et al. 2002, LaPorte et al. 2008) and IL2RG phosphorylation is induced only by IL4 (Roy et al. 2002).

Both IL4 receptor complexes signal through Jak/STAT cascades. IL4R is constitutively-associated with JAK2 (Roy et al. 2002) and associates with JAK1 following binding of IL4 (Yin et al. 1994) or IL13 (Roy et al. 2002). IL2RG constitutively associates with JAK3 (Boussiotis et al. 1994, Russell et al. 1994). IL13RA1 constitutively associates with TYK2 (Umeshita-Suyama et al. 2000, Roy et al. 2002, LaPorte et al. 2008, Bhattacharjee et al. 2013).

IL4 binding to IL4R1 leads to phosphorylation of JAK1 (but not JAK2) and STAT6 activation (Takeda et al. 1994, Ratthe et al. 2007, Bhattacharjee et al. 2013).

IL13 binding increases activating tyrosine-99 phosphorylation of IL13RA1 but not that of IL2RG. IL4 binding to IL2RG leads to its tyrosine phosphorylation (Roy et al. 2002). IL13 binding to IL4R2 leads to TYK2 and JAK2 (but not JAK1) phosphorylation (Roy & Cathcart 1998, Roy et al. 2002).

Phosphorylated TYK2 binds and phosphorylates STAT6 and possibly STAT1 (Bhattacharjee et al. 2013).

A second mechanism of signal transduction activated by IL4 and IL13 leads to the insulin receptor substrate (IRS) family (Kelly-Welch et al. 2003). IL4R1 associates with insulin receptor substrate 2 and activates the PI3K/Akt and Ras/MEK/Erk pathways involved in cell proliferation, survival and translational control. IL4R2 does not associate with insulin receptor substrate 2 and consequently the PI3K/Akt and Ras/MEK/Erk pathways are not activated (Busch-Dienstfertig & González-Rodríguez 2013).

References

- Nelms K, Keegan AD, Zamorano J, Ryan JJ & Paul WE (1999). The IL-4 receptor: signaling mechanisms and biologic functions. *Annu. Rev. Immunol.*, 17, 701-38. [↗](#)
- Hershey GK (2003). IL-13 receptors and signaling pathways: an evolving web. *J. Allergy Clin. Immunol.*, 111, 677-90; quiz 691. [↗](#)

Edit history

Date	Action	Author
2015-07-01	Authored	Jupe S
2015-07-01	Created	Jupe S
2016-09-02	Edited	Jupe S
2016-09-02	Reviewed	Leibovich SJ

17 submitted entities found in this pathway, mapping to 34 Reactome entities

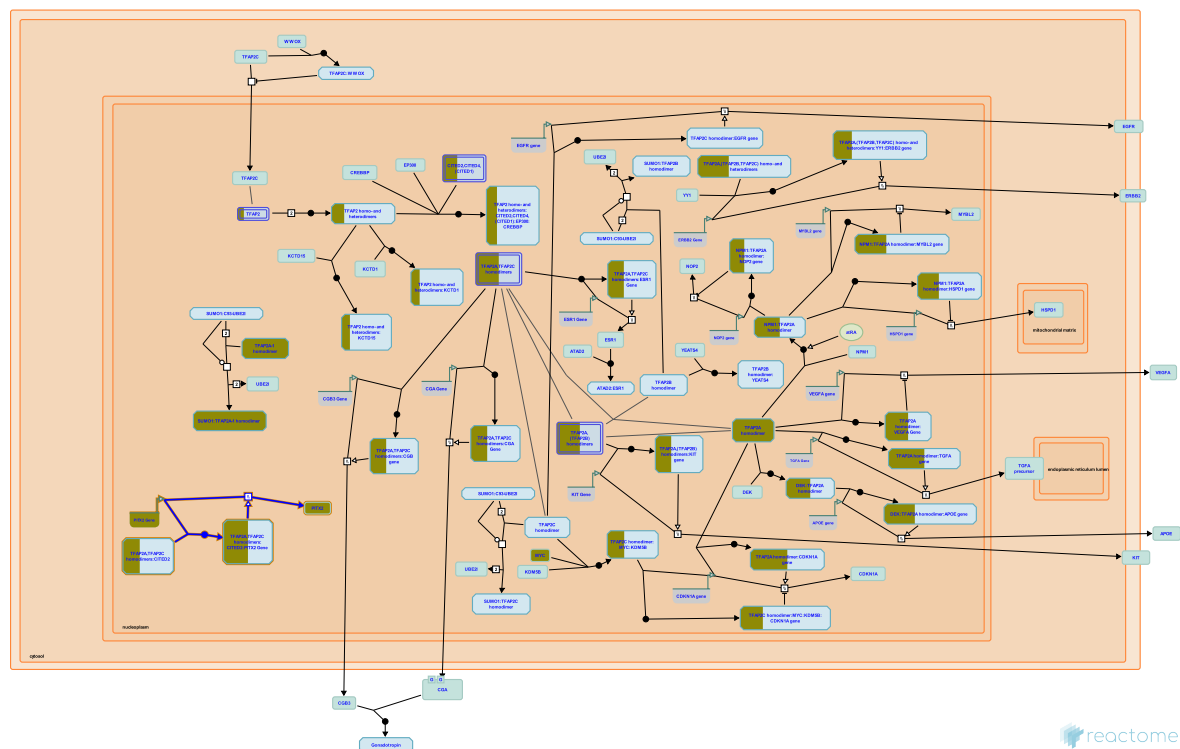
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BCL2L1	Q07817	CXCL8	P10145	FGF2	P09038
FOXO1	Q12778	FOXO3	O43524	HMOX1	P09601
ICAM1	P05362	IL1B	P01583, P01584	IL6	P05231
LIF	P15018	MCL1	Q07820	MYC	P01106
PIK3R1	P27986	S1PR1	P21453	SAA1	P0DJ18
SOCS5	O75159	ZEB1	P37275		

Input	Ensembl Id	Input	Ensembl Id	Input	Ensembl Id
BCL2L1	ENSG00000171552	CXCL8	ENSG00000169429	FGF2	ENSG00000138685
FOXO1	ENSG00000150907	FOXO3	ENSG00000118689	HMOX1	ENSG00000100292
ICAM1	ENSG00000090339	IL1B	ENSG00000125538	IL6	ENSG00000136244
LIF	ENSG00000128342	MCL1	ENSG00000143384	MYC	ENSG00000136997
PIK3R1	ENSG00000145675	S1PR1	ENSG00000170989	SAA1	ENSG00000173432
ZEB1	ENSG00000148516				

Interactors found in this pathway (11)

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
C5AR2	Q9P296	P09601	HIP1R	O75146	Q14627
IFNGR1	P15260	P23458	IL6ST	P40189-1	P23458
JAGN1	Q8N5M9	P09601	MYC	EBI-1265559	P23771
PIK3R1	P27986-2, P27986	P48023, O15524	SNX30	Q8WV41	P48023
TGFBR2	P37173	P23458	TNFSF9	P41273	Q14627
TRIP10	Q15642	P48023			

4. TFAP2 (AP-2) family regulates transcription of other transcription factors (R-HSA-8866906)



Homodimers and possibly heterodimers of TFAP2A and TFAP2C, in complex with CITED2, stimulate transcription of the PITX2 gene, involved in left-right patterning and heart development (Bamforth et al. 2004, Li et al. 2012).

References

Li Q, Pan H, Guan L, Su D & Ma X (2012). CITED2 mutation links congenital heart defects to dysregulation of the cardiac gene VEGF and PITX2C expression. Biochem. Biophys. Res. Commun., 423, 895-9. [🔗](#)

Bamforth SD, Bragança J, Farthing CR, Schneider JE, Broadbent C, Michell AC, ... Bhattacharya S (2004). Cited2 controls left-right patterning and heart development through a Nodal-Pitx2c pathway. Nat. Genet., 36, 1189-96. [🔗](#)

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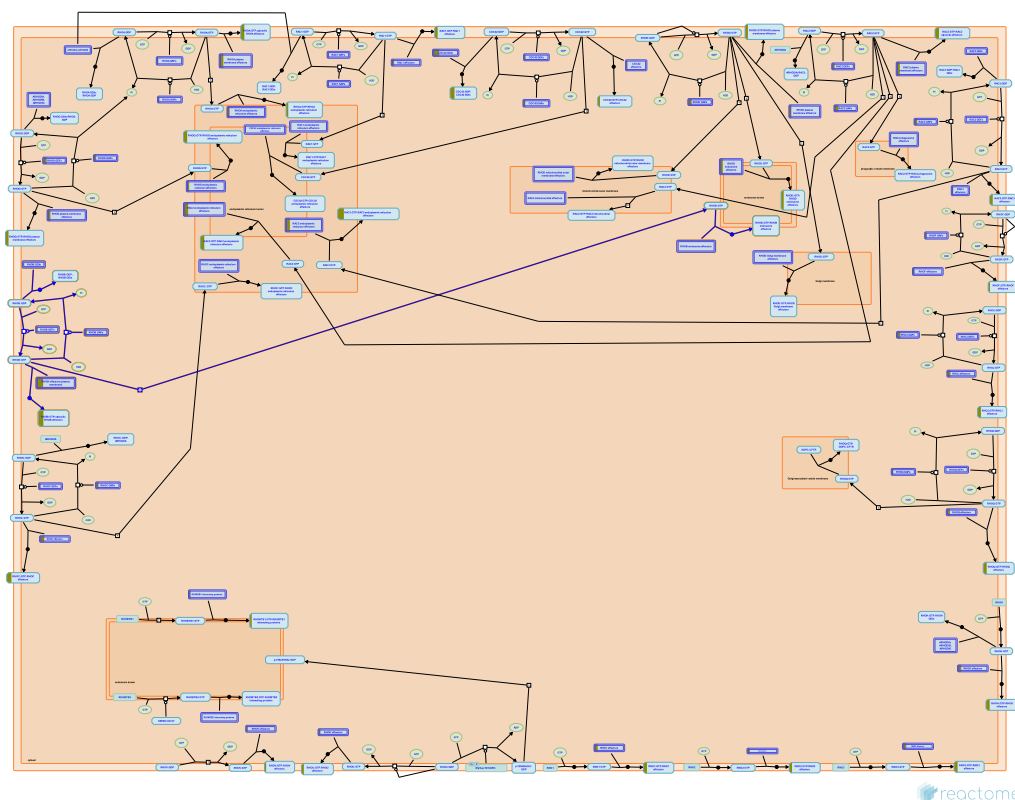
Date	Action	Author
2016-03-14	Edited	Orlic-Milacic M
2016-03-14	Authored	Orlic-Milacic M
2016-04-04	Created	Orlic-Milacic M
2016-05-04	Reviewed	Dawid IB, Zarelli VE
2016-05-17	Reviewed	Bogachek MV, Weigel RJ
2023-03-08	Modified	Matthews L

2 submitted entities found in this pathway, mapping to 3 Reactome entities

Input	UniProt Id	Input	UniProt Id
PITX2	Q99697	TFAP2A	P05549

Input	Ensembl Id
PITX2	ENSG00000164093

5. RHOB GTPase cycle (R-HSA-9013026)



This pathway catalogues RHOB guanine nucleotide exchange factors (GEFs), GTPase activator proteins (GAPs), GDP dissociation inhibitors (GDIs) and RHOB effectors. RHOB belongs to the RHOA subfamily of RHO GTPases and shares 85% sequence identity with RHOA and RHOC. Posttranslational modifications of the unique C-terminus of RHOB regulate its subcellular localization. Similar to RHOA and RHOC, RHOB regulates the cytoskeleton and plays a role in cell migration. RHOB plays a role in regulation of membrane trafficking (Vega and Ridley 2018), cell proliferation (Vega and Ridley 2018), cell adhesion (Vega and Ridley 2018), DNA repair (Vega and Ridley 2018), and apoptosis (Prendergast 2001; Vega and Ridley 2018). RHOB-regulated cellular processes are important for the immune system function (Vega and Ridley 2018), pulmonary gas exchange (Wojciak-Stothard et al. 2012; Vega and Ridley 2018), and angiogenesis and vascular function (Vega and Ridley 2018). RHOB has been implicated both as a tumor suppressor and as an oncogene in cancer (Prendergast 2001; Huang and Prendergast 2006; Ju and Gilkes 2018; Calvayrac et al. 2018).

References

- Ju JA & Gilkes DM (2018). RhoB: Team Oncogene or Team Tumor Suppressor?. *Genes (Basel)*, 9. [🔗](#)
- Vega FM & Ridley AJ (2018). The RhoB small GTPase in physiology and disease. *Small GTPases*, 9, 384-393. [🔗](#)
- Wojciak-Stothard B, Zhao L, Oliver E, Dubois O, Wu Y, Kardassis D, ... Wilkins MR (2012). Role of RhoB in the regulation of pulmonary endothelial and smooth muscle cell responses to hypoxia. *Circ. Res.*, 110, 1423-34. [🔗](#)
- Calvayrac O, Pradines A & Favre G (2018). RHOB expression controls the activity of serine/threonine protein phosphatase PP2A to modulate mesenchymal phenotype and invasion in non-small cell lung cancers. *Small GTPases*, 9, 339-344. [🔗](#)
- Huang M & Prendergast GC (2006). RhoB in cancer suppression. *Histol. Histopathol.*, 21, 213-8. [🔗](#)

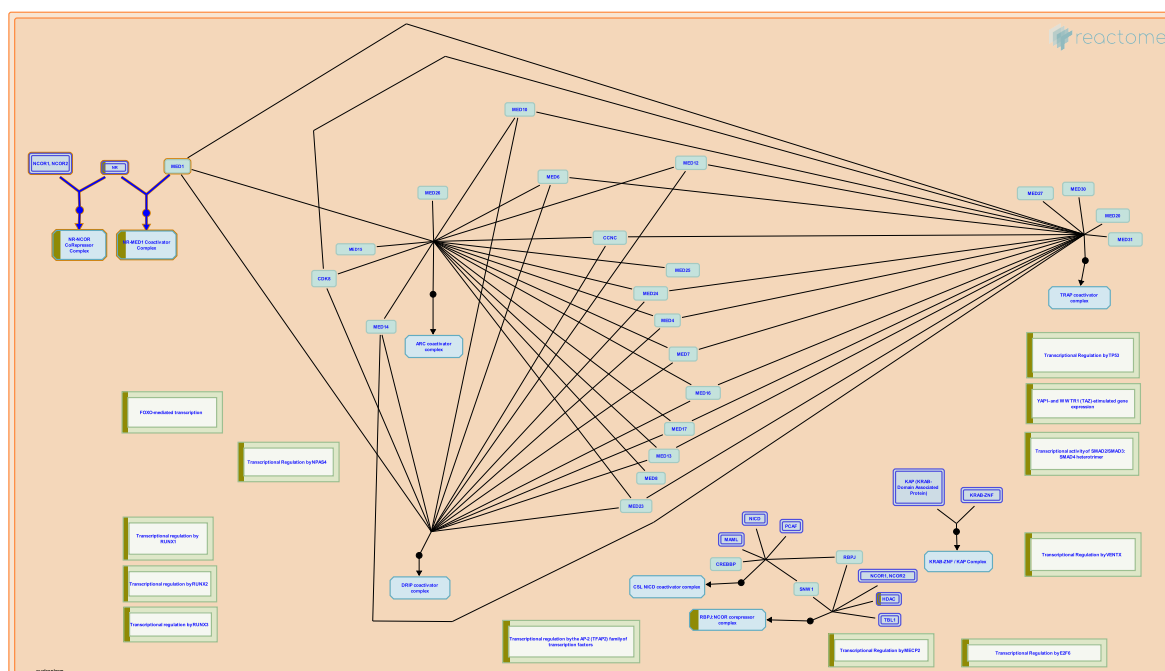
Edit history

Date	Action	Author
2017-07-21	Created	Orlic-Milacic M
2020-07-14	Authored	Orlic-Milacic M
2021-02-05	Reviewed	Fort P
2021-02-25	Edited	Orlic-Milacic M
2025-11-15	Modified	Weiser JD

9 submitted entities found in this pathway, mapping to 9 Reactome entities

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
AKAP13	Q12802	ARHGAP26	Q9UNA1	CAV1	Q03135
DAAM1	Q9Y4D1	MID1	Q9NQW6	PIK3R1	P27986
ROCK1	Q13464	RTKN	Q9BST9	TJP2	Q9UDY2

6. Nuclear Receptor transcription pathway (R-HSA-383280)



A classic example of bifunctional transcription factors is the family of Nuclear Receptor (NR) proteins. These are DNA-binding transcription factors that bind certain hormones, vitamins, and other small, diffusible signaling molecules. The non-liganded NRs recruit specific corepressor complexes of the NCOR/SMRT type, to mediate transcriptional repression of the target genes to which they are bound. During signaling, ligand binding to a specific domain the NR proteins induces a conformational change that results in the exchange of the associated CoR complex, and its replacement by a specific coactivator complex of the TRAP / DRIP / Mediator type. These coactivator complexes typically nucleate around a MED1 coactivator protein that is directly bound to the NR transcription factor.

A general feature of the 49 human NR proteins is that in the unliganded state, they each bind directly to an NCOR corepressor protein, either NCOR1 or NCOR2 (NCOR2 was previously named "SMRT"). This NCOR protein nucleates the assembly of additional, specific corepressor proteins, depending on the cell and DNA context. The NR-NCOR interaction is mediated by a specific protein interaction domain (PID) present in the NRs that binds to specific cognate PID(s) present in the NCOR proteins. Thus, the human NRs each take part in an NR-NCOR binding reaction in the absence of binding by their ligand.

A second general feature of the NR proteins is that they each contain an additional, but different PID that mediates specific binding interactions with MED1 proteins. In the ligand-bound state, NRs each take part in an NR-MED1 binding reaction to form an NR-MED1 complex. The bound MED1 then functions to nucleate the assembly of additional specific coactivator proteins, depending on the cell and DNA context, such as what specific target gene promoter they are bound to, and in what cell type.

The formation of specific MED1-containing coactivator complexes on specific NR proteins has been well-characterized for a number of the human NR proteins (see Table 1 in (Bourbon, 2004)). For example, binding of thyroid hormone (TH) to the human TH Receptor (THRA or THRB) was found to result in the recruitment of a specific complex of Thyroid Receptor Associated Proteins - the TRAP coactivator complex - of which the TRAP220 subunit was later identified to be the Mediator 1 (MED1) homologue.

Similarly, binding of Vitamin D to the human Vitamin D3 Receptor was found to result in the recruitment of a specific complex of D Receptor Interacting Proteins - the DRIP coactivator complex, of which the DRIP205 subunit was later identified to be human MED1.

References

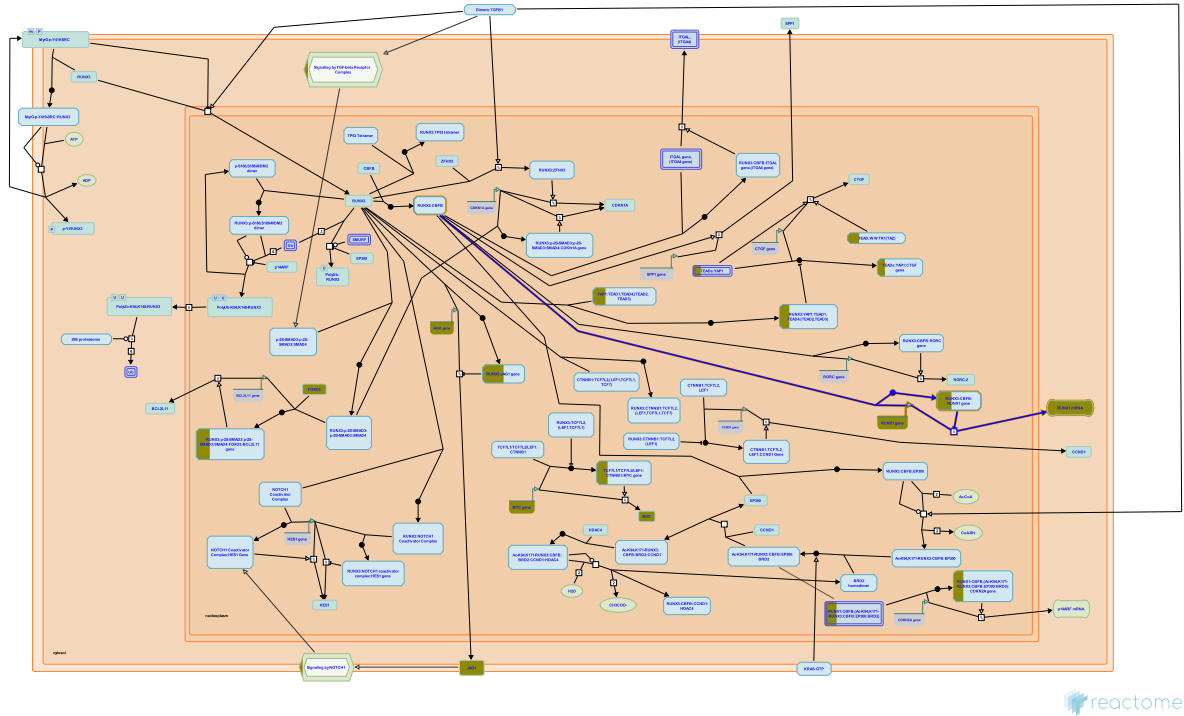
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Date	Action	Author
2008-11-20	Authored	Caudy M
2008-12-03	Created	Caudy M
2009-05-27	Edited	Caudy M
2009-08-29	Reviewed	Freedman LP

4 submitted entities found in this pathway, mapping to 12 Reactome entities

Input	UniProt Id	Input	UniProt Id
NR2F1	P10589	NR3C1	P04150-1, P04150-2, P04150-3, P04150-4, P04150-5, P04150-6, P04150-7, P04150-8, P04150-9
RARG	P13631	THRB	P10828

7. RUNX3 regulates RUNX1-mediated transcription (R-HSA-8951911)



RUNX3 binds to Runx response elements in the distal (P1) promoter of the RUNX1 gene, repressing RUNX1 transcription (Spender et al. 2005).

References

Spender LC, Whiteman HJ, Karstegl CE & Farrell PJ (2005). Transcriptional cross-regulation of RUNX1 by RUNX3 in human B cells. *Oncogene*, 24, 1873-81. [🔗](#)

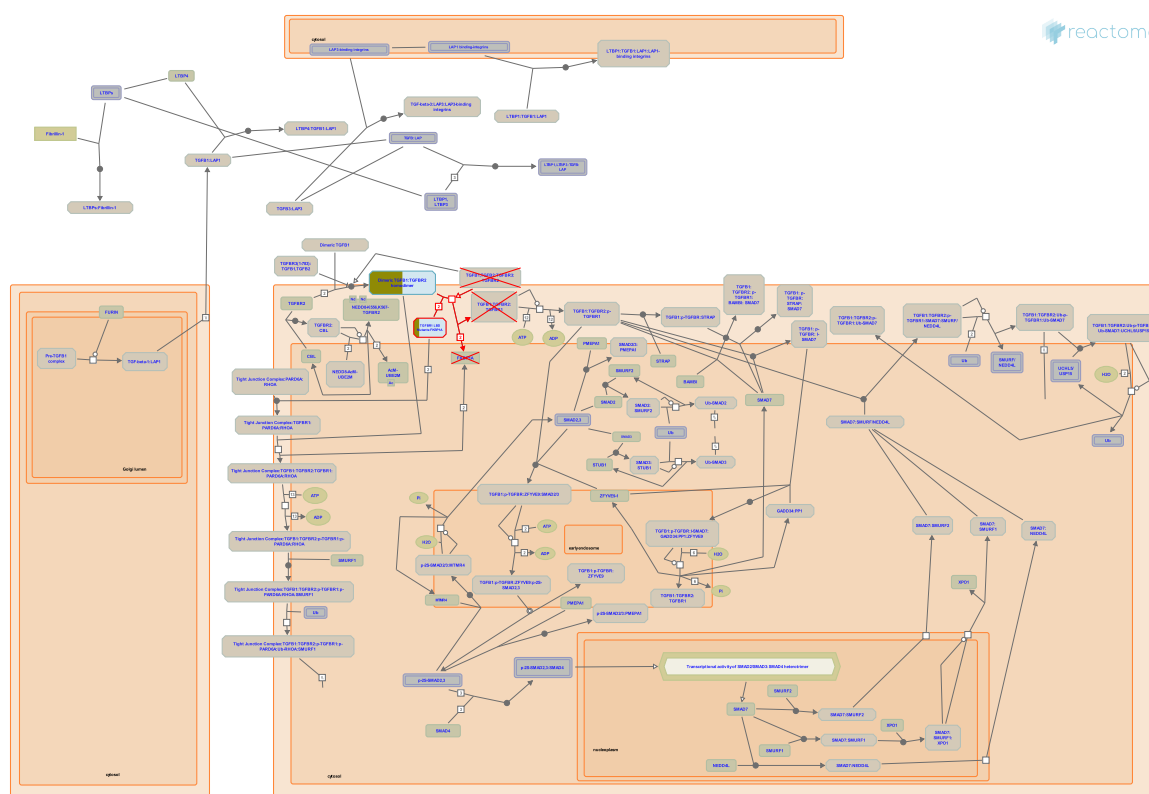
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Date	Action	Author
2016-12-12	Created	Orlic-Milacic M
2016-12-13	Authored	Orlic-Milacic M
2017-01-31	Edited	Orlic-Milacic M
2017-01-31	Reviewed	Ito Y, Chuang LS
2023-03-08	Modified	Matthews L

1 submitted entities found in this pathway, mapping to 2 Reactome entities

Input	Ensembl Id
RUNX1	ENSG00000159216, ENST00000344691

8. TGFBR1 LBD Mutants in Cancer (R-HSA-3656535)



Diseases: cancer.

Mutations in the ligand-binding domain (LBD) of TGF-beta receptor 1 (TGFBR1) have been reported as germline mutations in Ferguson-Smith tumor (multiple self-healing squamous epithelioma - MSSE), an autosomal-dominant skin cancer condition (Ferguson-Smith et al. 1934, Ferguson-Smith et al. 1971), with tumors frequently showing loss of heterozygosity of the wild-type TGFBR1 allele (Goudie et al. 2011). Somatic mutations in the LBD of TGFBR1 have been reported in esophageal carcinoma (Dulak et al. 2013).

References

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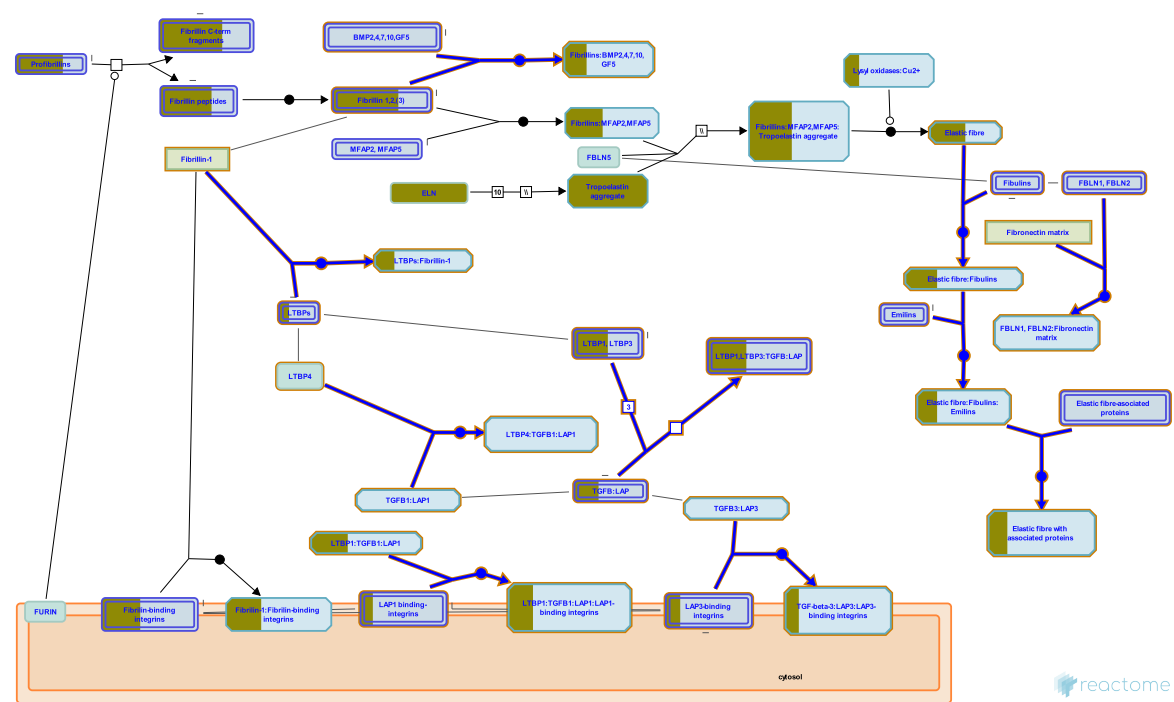
Date	Action	Author
2013-06-05	Created	Orlic-Milacic M

Date	Action	Author
2013-08-08	Edited	Orlic-Milacic M
2013-08-08	Reviewed	Meyer S, Akhurst RJ
2013-08-08	Authored	Meyer S, Akhurst RJ, Orlic-Milacic M

2 submitted entities found in this pathway, mapping to 2 Reactome entities

Input	UniProt Id	Input	UniProt Id
FPR1	P62942	TGFBR2	P37173

9. Molecules associated with elastic fibres (R-HSA-2129379)



Proteins found associated with microfibrils include vitronectin (Dahlback et al. 1990), latent transforming growth factor beta-binding proteins (Kielty et al. 2002, Munger & Sheppard 2011), emilin (Bressan et al. 1993, Mongiat et al. 2000), members of the microfibrillar-associated proteins (MFAPs, Gibson et al.1996), and fibulins (Roark et al. 1995, Yanagisawa et al. 2002). The significance of these interactions is not well understood but may help mediate elastin-fibrillin interactions during elastic fibre assembly.

Proteoglycans such as versican (Isogai et al. 2002), biglycan, and decorin (Reinboth et al. 2002) can interact with the microfibrils. They confer specific properties including hydration, impact absorption, molecular sieving, regulation of cellular activities, mediation of growth factor association, and release and transport within the extracellular matrix (Buczek-Thomas et al. 2002). In addition, glycosaminoglycans have been shown to interact with tropoelastin through its lysine side chains (Wu et al. 1999) regulating tropoelastin assembly (Tu and Weiss, 2008).

References

Wagenseil JE & Mecham RP (2007). New insights into elastic fiber assembly. Birth Defects Res. C Embryo Today, 81, 229-40. [🔗](#)

Kielty CM, Sherratt MJ & Shuttleworth CA (2002). Elastic fibres. J Cell Sci, 115, 2817-28. [🔗](#)

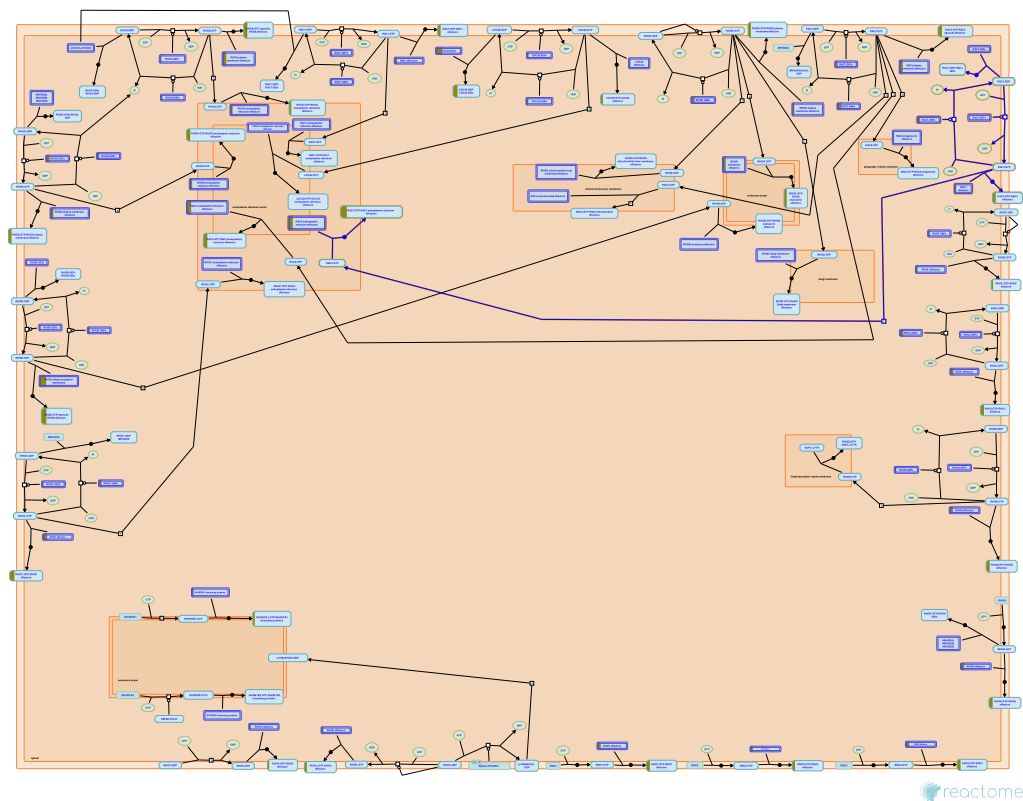
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Date	Action	Author
2012-02-21	Created	Jupe S
2012-04-30	Authored	Jupe S
2012-11-02	Reviewed	Muiznieks LD
2012-11-12	Edited	Jupe S
2025-11-15	Modified	Weiser JD

5 submitted entities found in this pathway, mapping to 6 Reactome entities

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
ELN	P15502	FBN2	P35556, Q75N90	ITGB3	P05106
LTBP1	Q14766	TGFB2	P61812		

10. RAC3 GTPase cycle (R-HSA-9013423)



This pathway catalogues RAC3 guanine nucleotide exchange factors (GEFs), GTPase activator proteins (GAPs), GDP dissociation inhibitors (GDIs) and RAC3 effectors. RAC3 is highly similar to RAC1 (92% amino acid sequence identity), but it is expressed in fewer tissues than RAC1. RAC3 orthologues only exist in vertebrates (de Curtis 2019). RAC3 is highly expressed in neurons and plays an important role in neuronal and brain development (de Curtis 2014, de Curtis 2019). RAC3 mutations have been reported in patients with intellectual disability and brain malformations (de Curtis 2019). RAC3 is frequently overexpressed in cancer and contributes to proliferation, migration and invasiveness of cancer cells (de Curtis 2019).

References

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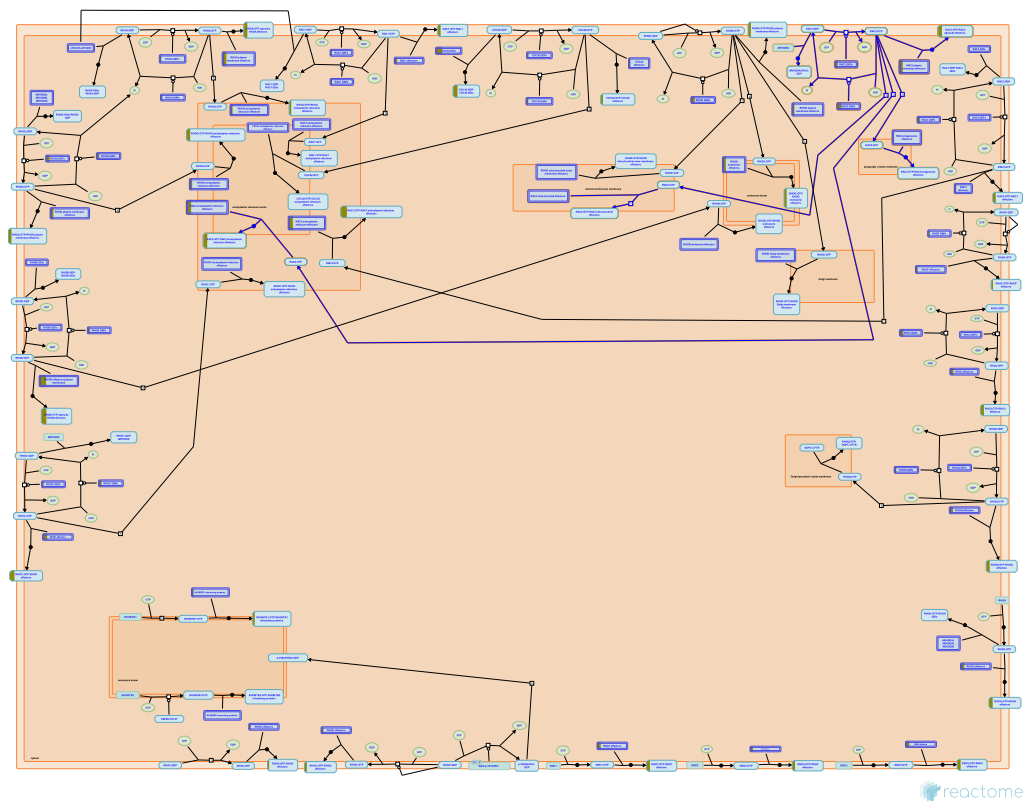
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Date	Action	Author
2017-07-25	Created	Orlic-Milacic M
2020-07-14	Authored	Rothfels K, Orlic-Milacic M
2021-02-05	Reviewed	Fort P
2021-02-25	Edited	Orlic-Milacic M
2025-11-15	Modified	Weiser JD

11 submitted entities found in this pathway, mapping to 11 Reactome entities

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
ARHGAP26	Q9UNA1	ARHGAP42	A6NI28	BAIAP2L1	Q9UHR4
CAV1	Q03135	EPHA2	P29317	JAG1	P78504
NHS	Q6T4R5	PGRMC2	O15173	PIK3R1	P27986
SLC1A5	Q15758	TMEM133	A6NI28		

11. RAC2 GTPase cycle (R-HSA-9013404)



This pathway catalogues RAC2 guanine nucleotide exchange factors (GEFs), GTPase activator proteins (GAPs), GDP dissociation inhibitors (GDIs) and RAC2 effectors. RAC2 is exclusively expressed in hematopoietic cells (Troeger and Williams 2013). RAC2 is a component of the phagocytic oxidase complex in neutrophils (Troeger and Williams 2013). RAC2 is required for adhesion and mobilization of hematopoietic stem cells and progenitor cells (Troeger and Williams 2013). RAC2 is also needed for adhesion, migration and degranulation of mast cells (Troeger and Williams 2013). Mutations in RAC2 have been found in a small number of patients with primary immunodeficiencies (Gu and Williams 2002; Troeger and Williams 2013; Lougaris et al. 2020).

References

Lougaris V, Baronio M, Gazzurelli L, Benvenuto A & Plebani A (2020). RAC2 and primary human immune deficiencies. *J. Leukoc. Biol.* [🔗](#)

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Gu Y & Williams DA (2002). RAC2 GTPase deficiency and myeloid cell dysfunction in human and mouse. *J. Pediatr. Hematol. Oncol.*, 24, 791-4. [🔗](#)

Edit history

Date	Action	Author
2017-07-25	Created	Orlic-Milacic M
2020-07-14	Authored	Rothfels K, Orlic-Milacic M
2021-02-05	Reviewed	Fort P
2021-02-25	Edited	Orlic-Milacic M

Date	Action	Author
2025-11-15	Modified	Weiser JD

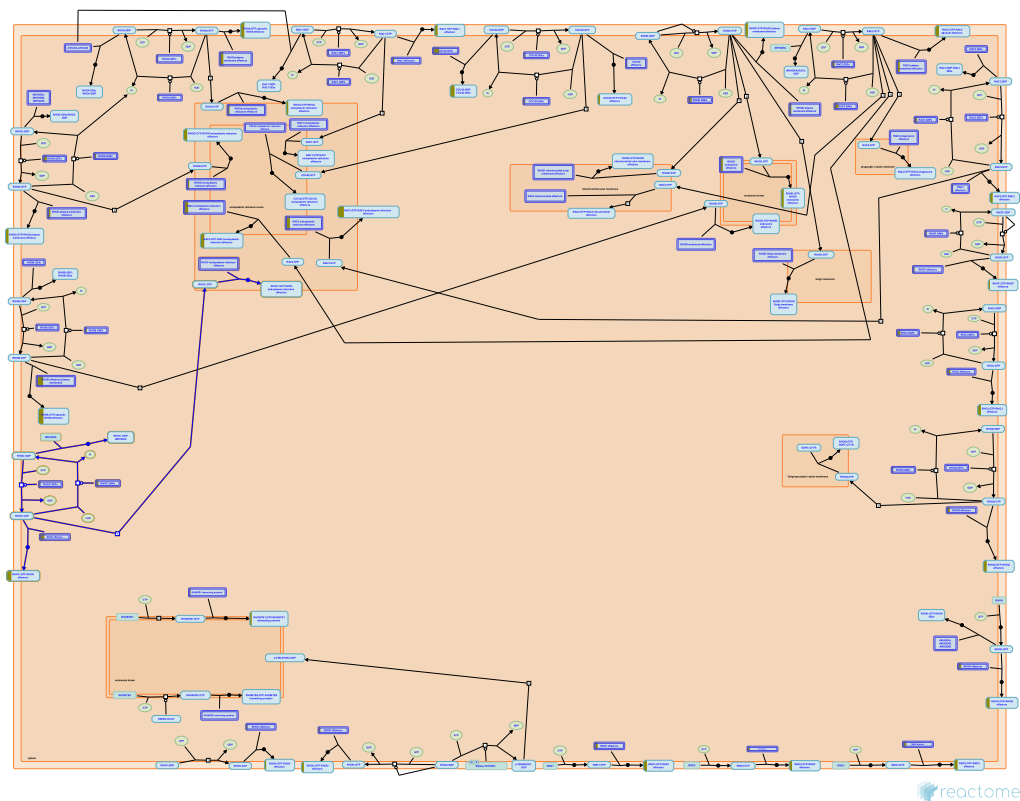
11 submitted entities found in this pathway, mapping to 11 Reactome entities

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ARHGAP26	Q9UNA1	ARHGAP42	A6NI28	BAIAP2L1	Q9UHR4
CAV1	Q03135	DOCK4	Q8N1I0	EPHA2	P29317
NHS	Q6T4R5	PGRMC2	O15173	PIK3R1	P27986
TMEM133	A6NI28	VAV3	Q9UKW4		

Interactors found in this pathway (1)

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
TNFRSF19	Q9NS68	P52565			

12. RHOC GTPase cycle (R-HSA-9013106)



This pathway catalogues RHOC guanine nucleotide exchange factors (GEFs), GTPase activator proteins (GAPs), GDP dissociation inhibitors (GDIs) and RHOC effectors. RHOC belongs to the RHO subfamily of RHO GTPases and shares 85% sequence identity with RHOA and RHOB (Wheeler and Ridley 2004). Like RHOA and RHOB, RHOC regulates the cytoskeleton and is involved in cell adhesion and migration (Guan et al. 2018). RHOC contributes to invasiveness and metastatic potential of cancer cells (Bravo-Cordero et al. 2014; Guan et al. 2018; Thomas et al. 2019).

References

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Edit history

Date	Action	Author
2017-07-23	Created	Orlic-Milacic M
2020-07-14	Authored	Orlic-Milacic M
2021-02-05	Reviewed	Fort P

Date	Action	Author
2021-02-25	Edited	Orlic-Milacic M
2025-11-15	Modified	Weiser JD

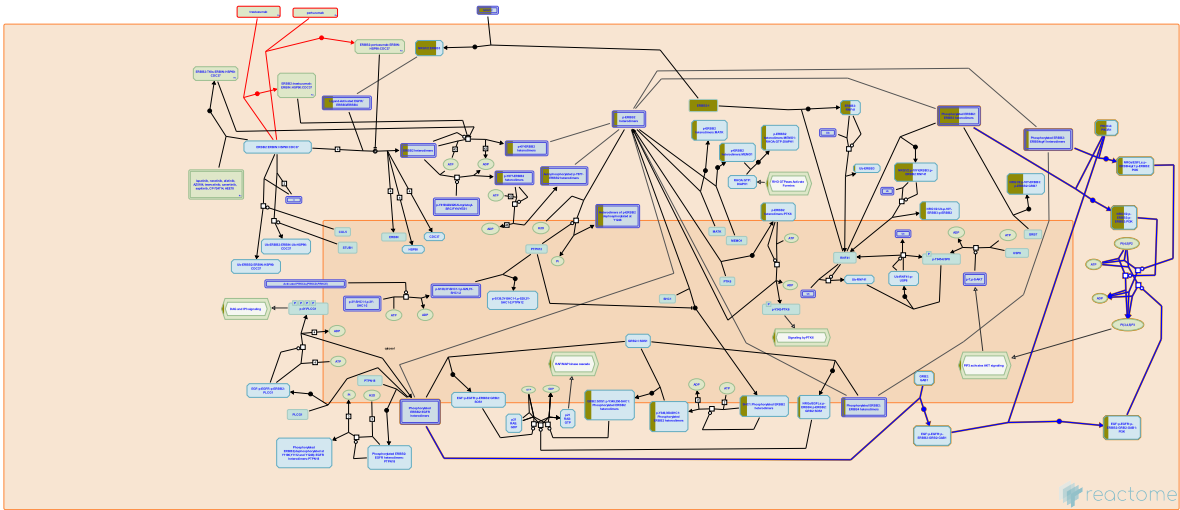
9 submitted entities found in this pathway, mapping to 9 Reactome entities

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AKAP13	Q12802	ARHGAP26	Q9UNA1	CAV1	Q03135
DAAM1	Q9Y4D1	MID1	Q9NQW6	PIK3R1	P27986
ROCK1	Q13464	RTKN	Q9BST9	TJP2	Q9UDY2

Interactors found in this pathway (1)

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
TNFRSF19	Q9NS68	P52565			

13. PI3K events in ERBB2 signaling (R-HSA-1963642)



ERBB2:ERBB3 and ERBB2:ERBB4cyt1 heterodimers activate PI3K signaling by direct binding of PI3K regulatory subunit p85 (Yang et al. 2007, Cohen et al. 1996, Kaushansky et al. 2008) to phosphorylated tyrosine residues in the C-tail of ERBB3 (Y1054, Y1197, Y1222, Y1224, Y1276 and Y1289) and ERBB4 CYT1 isoforms (Y1056 in JM-A CYT1 isoform and Y1046 in JM-B CYT1 isoform). Regulatory subunit p85 subsequently recruits catalytic subunit p110 of PI3K, resulting in the formation of active PI3K, conversion of PIP2 to PIP3, and PIP3-mediated activation of AKT signaling (Junttila et al. 2009, Kainulainen et al. 2000). Heterodimers of ERBB2 and EGFR recruit PI3K indirectly, through GRB2:GAB1 complex (Jackson et al. 2004), which again leads to PIP3-mediated activation of AKT signaling.

References

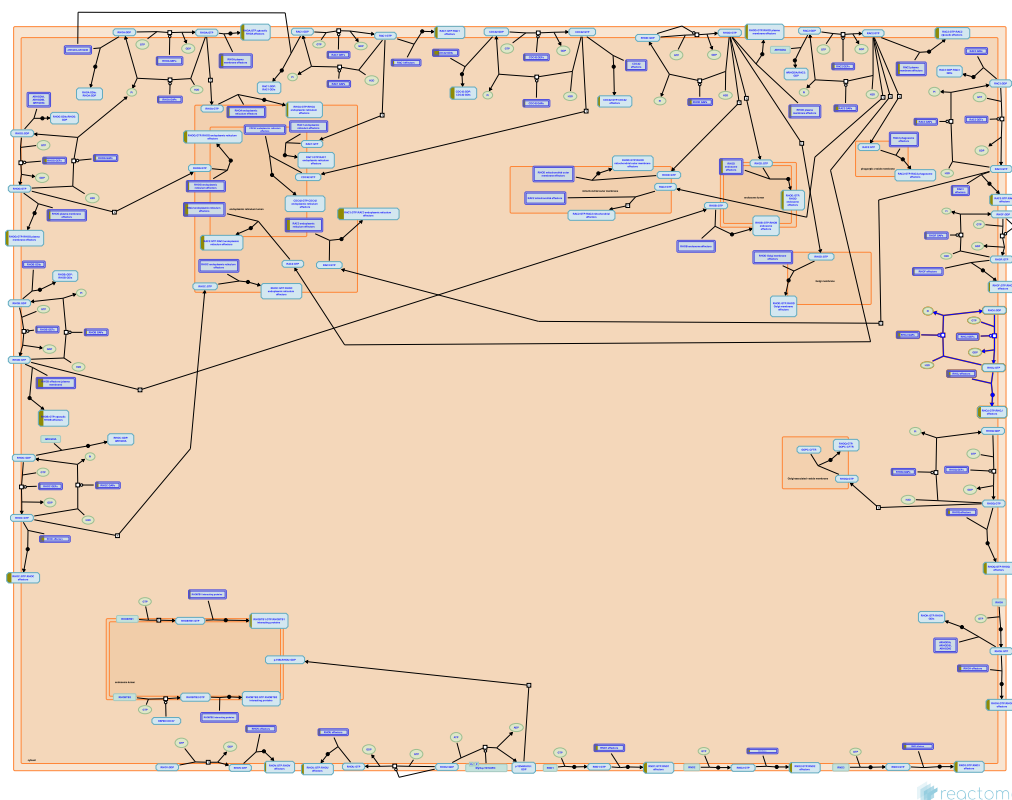
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Date	Action	Author
2011-11-04	Authored	Orlic-Milacic M
2011-11-04	Created	Orlic-Milacic M
2011-11-07	Edited	Matthews L, D'Eustachio P
2011-11-11	Reviewed	Xu W, Neckers LM

3 submitted entities found in this pathway, mapping to 4 Reactome entities

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
ERBB3	P21860-1	NRG1	Q02297, Q02297-10	PIK3R1	P27986

14. RHOJ GTPase cycle (R-HSA-9013409)



This pathway catalogues RHOJ (also known as TCL or RHOI) guanine nucleotide exchange factors (GEFs), GTPase activator proteins (GAPs), and RHOJ effectors. No GDP dissociation inhibitors (GDIs) have been shown to interact with RHOJ. RHOJ, together with RHOQ (TC10), belongs to the CDC42 subfamily of RHO GTPases and shares 85% and 78% amino acid similarity with RHOQ and CDC42, respectively (Vignal et al. 2000). RHOJ regulates the cytoskeleton, including formation of lamellipodia and actin filaments (Vignal et al. 2000, Abe et al. 2003, Aspenstrom et al. 2004, Shi et al. 2016). RHOJ is highly expressed in endothelial cells, regulating their motility and, consequently, vascular morphogenesis (Leszczynska et al. 2011; Shi et al. 2016). As a part of the VEGF signaling cascade, RHOJ promotes retinal angiogenesis (Fukushima et al. 2013). RHOJ plays a role in cancer cell migration and cancer-related angiogenesis (Shi et al. 2016). RHOJ is involved in adipocyte differentiation (Nishizuka 2003; Shi et al. 2016).

References

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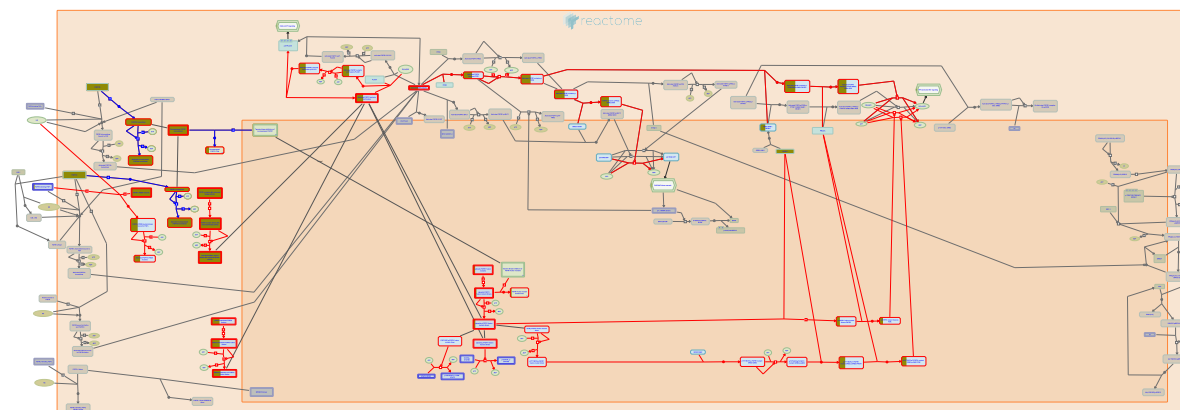
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2017-07-25	Created	Orlic-Milacic M
2020-07-14	Authored	Orlic-Milacic M
2021-02-05	Reviewed	Fort P
2021-02-25	Edited	Orlic-Milacic M
2021-05-17	Reviewed	Das ME
2021-06-10	Revised	Orlic-Milacic M
2021-06-10	Edited	Orlic-Milacic M
2025-11-15	Modified	Weiser JD

7 submitted entities found in this pathway, mapping to 7 Reactome entities

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
ARHGAP26	Q9UNA1	CAV1	Q03135	FNBP1L	Q5T0N5
GBAS	O75323	GJA1	P17302	PIK3R1	P27986
SLC1A5	Q15758				

15. Signaling by FGFR1 amplification mutants ([R-HSA-1839120](#))



Cellular compartments: cytosol, extracellular region, plasma membrane, nucleoplasm.

Diseases: cancer.

Amplification or activation of FGFR1 has been reported in lung cancer (Weiss, 2001; Marek, 2009; Dutt, 2011), breast cancer (Reis-Filho, 2006; Turner, 2010), oral squamous carcinoma (Freier, 2007), esophageal squamous cell carcinomas (Ishizuka, 2002), ovarian cancer (Gorringe, 2007), bladder cancer (Simon, 2001), prostate cancer (Edwards, 2003; Acevedo, 2007) and rhabdomyosarcoma (Missiaglia, 2009). Unlike the case for FGFR2 amplifications, FGFR1 amplifications are not associated with additional point mutations and affect signaling without altering the intrinsic kinase activity of the receptor. Overexpressed FGFR1 appears to signal at a basal level in a ligand-independent fashion, but is also able to be stimulated by exogenous ligand. Downstream activation may be the result of aberrant paracrine or autocrine stimulation (reviewed in Turner and Gross, 2010; Greulich and Pollock, 2011). FGFR1 amplification has not been conclusively demonstrated to be the causative oncogenic agent in all of the cancer types mentioned above, and other genes in the 8p11 region may also be candidates in some cases (Bass, 2009; Bernard-Pierrot, 2008; Ray, 2004).

References

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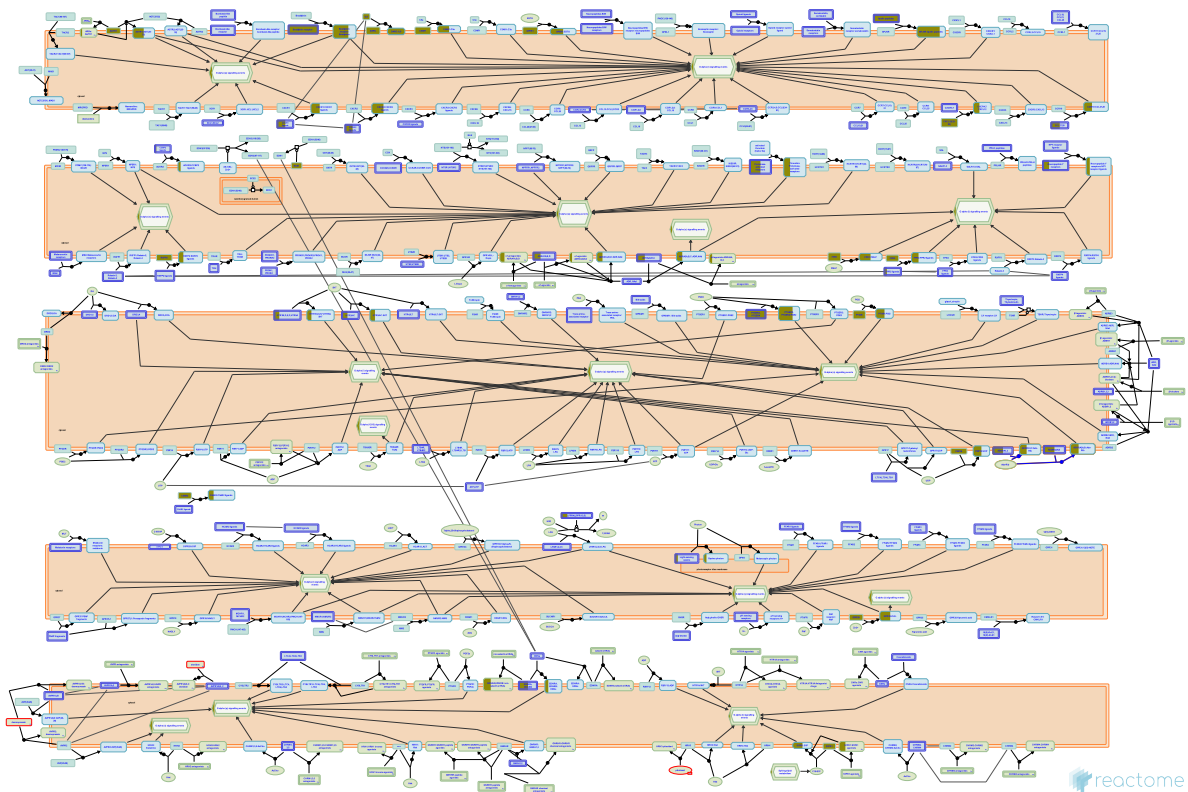
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Date	Action	Author
2011-10-27	Created	Rothfels K
2012-02-10	Authored	Rothfels K
2012-05-15	Reviewed	Ezzat S
2012-05-16	Edited	Rothfels K

1 submitted entities found in this pathway, mapping to 2 Reactome entities

Input	UniProt Id
FLG	P11362-1, P11362-19

16. Adenosine P1 receptors (R-HSA-417973)



The adenosine receptors (P1 receptors) are a class of purinergic receptors, G-protein coupled receptors with adenosine as their endogenous ligand. In humans, there are four adenosine receptors. Each is encoded by a separate gene and the four receptors have distinct, though overlapping, functions. For instance, both A1 and A2A receptors play roles in the heart, regulating myocardial oxygen consumption and coronary blood flow. They also have important roles in the brain, regulating the release of other neurotransmitters such as dopamine and glutamate. The A2B and A3 receptors are located peripherally and are involved in processes such as inflammation and immune responses. Fredholm BB et al, 2001).

References

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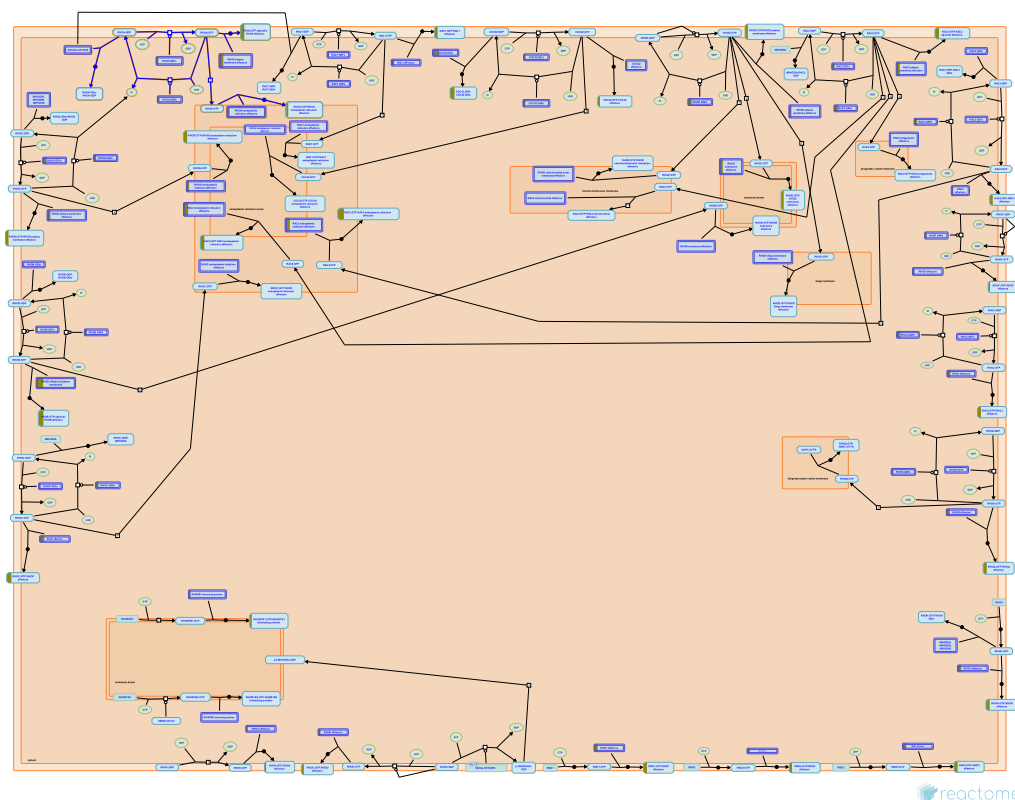
Edit history

Date	Action	Author
2009-04-14	Edited	Jassal B
2009-04-14	Authored	Jassal B
2009-04-14	Created	Jassal B
2009-05-29	Reviewed	D'Eustachio P
2025-11-15	Modified	Weiser JD

2 submitted entities found in this pathway, mapping to 2 Reactome entities

Input	UniProt Id	Input	UniProt Id
ADORA1	P30542	ADORA2B	P29275

17. RHOA GTPase cycle (R-HSA-8980692)



This pathway catalogues RHOA guanine nucleotide exchange factors (GEFs), GTPase activator proteins (GAPs), GDP dissociation inhibitors (GDIs) and RHOA effectors. RHOA is one of the three best characterized RHO GTPases, the other two being RAC1 and CDC42, and is the founding member of the RHO GTPase family (Zhou and Zheng 2013). RHOA regulates the cytoskeleton and cell contractility (Lessey et al. 2012), thus playing a role in a number of cellular functions, such as adhesion, migration, survival, division, vesicle trafficking and gene expression (Zhou and Zheng 2013). RHOA-regulated processes are involved in mechanotransduction (Lessey et al. 2012; Marjoram et al. 2014), neuronal development (Zhou and Zheng 2013; Fujita and Yamashita 2014; Hu and Selzer 2017), immune system development (Zhou and Zheng 2013; Ricker et al. 2016; Bros et al. 2019), and cardiovascular regulation (Zhou and Zheng 2013; Cai et al. 2016; Shimokawa et al. 2016). RHOA mutations are frequently found in cancer (Kataoka and Ogawa 2016). Toxins of numerous pathogens target RHOA to hijack the host cytoskeleton (Jamilloux et al. 2018).

References

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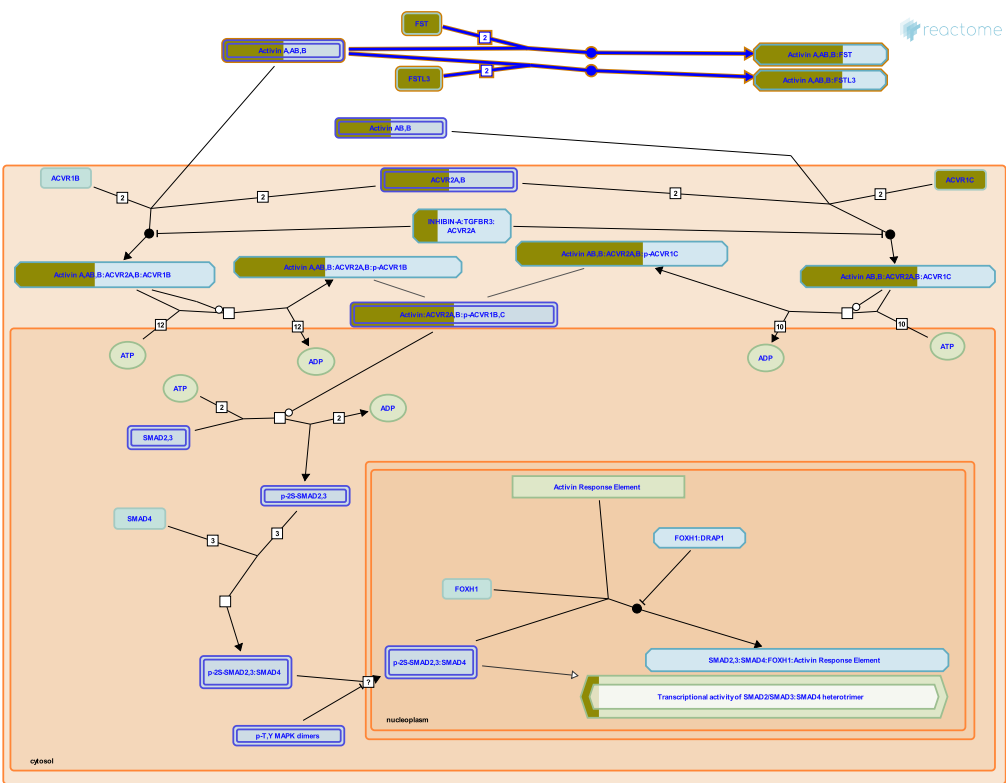
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Date	Action	Author
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2020-07-14	Authored	Orlic-Milacic M
2021-02-05	Reviewed	Fort P
2021-02-25	Edited	Orlic-Milacic M

14 submitted entities found in this pathway, mapping to 14 Reactome entities

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
AKAP13	Q12802	ARHGAP26	Q9UNA1	ARHGAP42	A6NI28
CAV1	Q03135	DAAM1	Q9Y4D1	KALRN	O60229
MID1	Q9NQW6	PGRMC2	O15173	PIK3R1	P27986
ROCK1	Q13464	RTKN	Q9BST9	TJP2	Q9UDY2
TMEM133	A6NI28	VAV3	Q9UKW4		

18. Antagonism of Activin by Follistatin (R-HSA-2473224)



Cellular compartments: extracellular region.

Both Follistatin (FST) and Follistatin-like-3 (FSTL3) irreversibly bind Activin dimers and prevent Activin from interacting with its receptor (reviewed in Schneyer et al. 2004, Xia and Schneyer 2009). Though functionally similar in vitro, FST and FSTL3 do not function identically in vivo. Mice lacking FST die shortly after birth due to defects in muscle and bone (Matzuk et al. 1995); mice lacking FSTL3 are viable but have altered glucose metabolism (Mukherjee et al. 2007).

References

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Xia Y & Schneyer AL (2009). The biology of activin: recent advances in structure, regulation and function. *J. Endocrinol.*, 202, 1-12. [🔗](#)

Matzuk MM, Lu N, Vogel H, Sellheyer K, Roop DR & Bradley A (1995). Multiple defects and perinatal death in mice deficient in follistatin. *Nature*, 374, 360-3. [🔗](#)

Mukherjee A, Sidis Y, Mahan A, Raher MJ, Xia Y, Rosen ED, ... Schneyer AL (2007). FSTL3 deletion reveals roles for TGF-beta family ligands in glucose and fat homeostasis in adults. *Proc. Natl. Acad. Sci. U.S.A.*, 104, 1348-53. [🔗](#)

Edit history

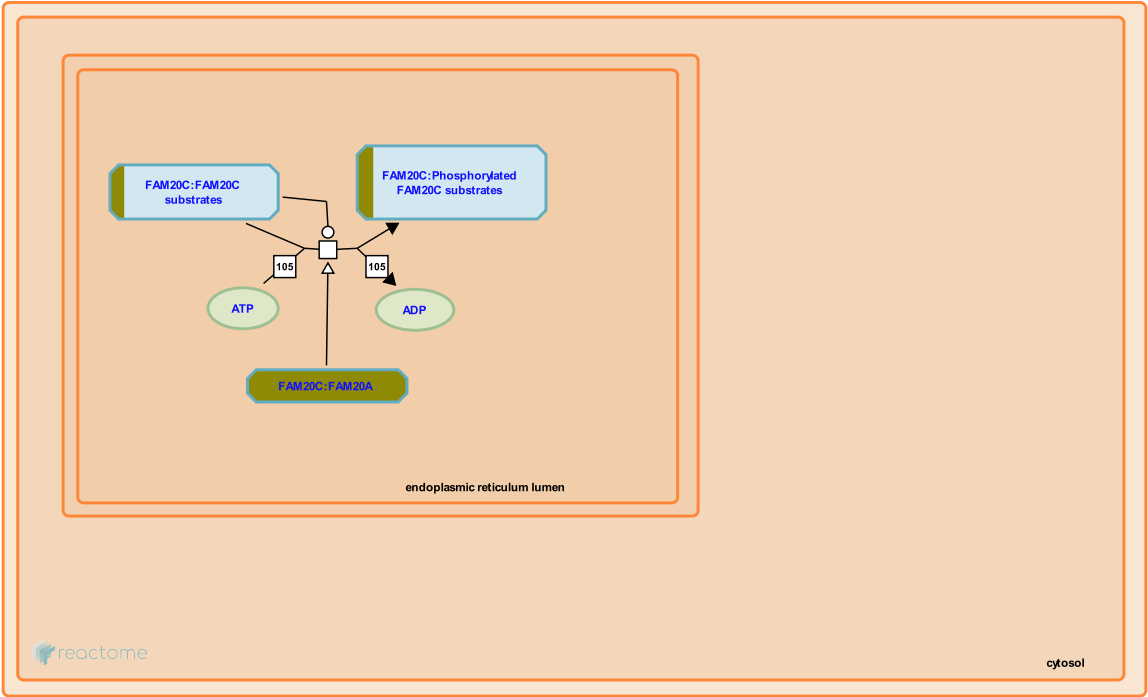
Date	Action	Author
2012-09-21	Edited	May B
2012-09-21	Authored	May B

Date	Action	Author
2012-09-22	Created	May B
2012-11-14	Reviewed	Chen YG
2025-11-15	Modified	Weiser JD

3 submitted entities found in this pathway, mapping to 3 Reactome entities

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
FST	P19883	FSTL3	O95633	INHBB	P09529

19. Post-translational protein phosphorylation (R-HSA-8957275)



Secretory pathway kinases phosphorylate a diverse array of substrates involved in many physiological processes.

References

Sreelatha A, Kinch LN & Tagliabracci VS (2015). The secretory pathway kinases. Biochim. Biophys. Acta, 1854, 1687-93. [↗](#)

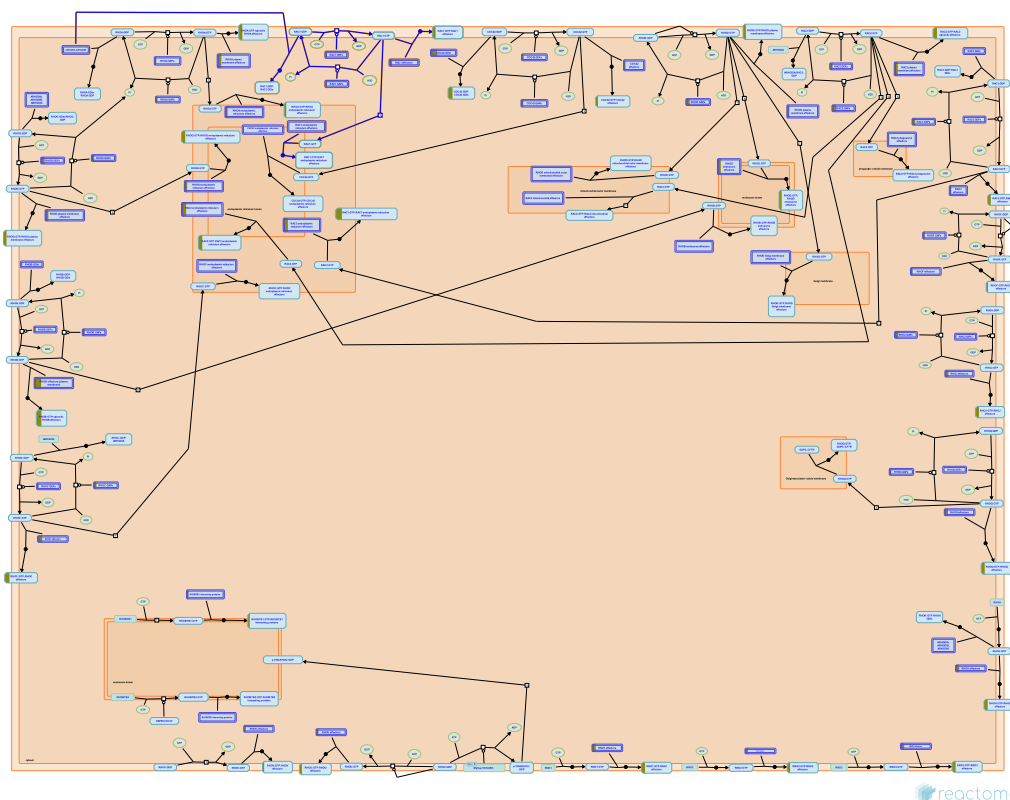
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Date	Action	Author
2016-12-08	Authored	Jupe S
2017-01-23	Reviewed	Wiley SE
2017-01-24	Edited	Jupe S
2017-01-24	Created	Jupe S

10 submitted entities found in this pathway, mapping to 10 Reactome entities

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
APOB	P04114	CSF1	P09603	FAM20A	Q96MK3
FAM20C	Q8IXL6	FSTL3	O95633	IL6	P05231
LTBP1	Q14766	PEAR1	P11047	PNPLA2	Q96AD5
STC2	O76061				

20. RAC1 GTPase cycle (R-HSA-9013149)



This pathway catalogues RAC1 guanine nucleotide exchange factors (GEFs), GTPase activator proteins (GAPs), GDP dissociation inhibitors (GDIs) and RAC1 effectors (reviewed by Payapilli and Malliri 2018). RAC1 is one of the three best characterized RHO GTPases, the other two being RHOA and CDC42. RAC1 regulates the cytoskeleton and the production of reactive oxygen species (ROS) (Acevedo and Gonzalez-Billault 2018) and is involved in cell adhesion and cell migration (Marei and Malliri 2017). RAC1 is involved in neuronal development (de Curtis et al. 2014). In neurons, RAC1 activity is regulated by synaptic activation and RAC1-mediated changes in actin cytoskeleton are implicated in dendritic spine morphogenesis, which plays a role in memory formation and learning (Tajeda-Simon 2015; Costa et al. 2020). RAC1 is involved in metabolic regulation of pancreatic islet β -cells and in diabetes pathophysiology (Kowluru 2017; Kowluru et al. 2020). RAC1-mediated activation of NOX2 contributes to mitochondrial damage and the development of retinopathy in patients with diabetes (Sahajpal et al. 2019). RAC1 is important for exercise and contraction-stimulated glucose uptake in skeletal muscles (Sylov et al. 2014). RAC1 plays an important role in the maintenance of intestinal barrier integrity under physiological conditions and during tissue repair after resolution of colitis. Toxins of many diarrhea-causing bacteria target RAC1 (Kotelevets and Chastre 2020). RAC1 is important for skin homeostasis and wound healing and is involved in the pathogenesis of psoriasis (Winge and Marinkovich 2019). RAC1 is essential to vascular homeostasis and chronically elevated RAC1 signaling contributes to vascular pathology (Marinkovic et al. 2015). RAC1 hyperactivation, mutation and copy-number gain are frequently observed in solid tumors (Zou et al. 2017; De et al. 2019; De et al. 2020; Cannon et al. 2020; Kotelevets and Chastre 2020).

References

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De P, Rozeboom BJ, Aske JC & Dey N (2020). Active RAC1 Promotes Tumorigenic Phenotypes and Therapy Resistance in Solid Tumors. *Cancers (Basel)*, 12. [🔗](#)

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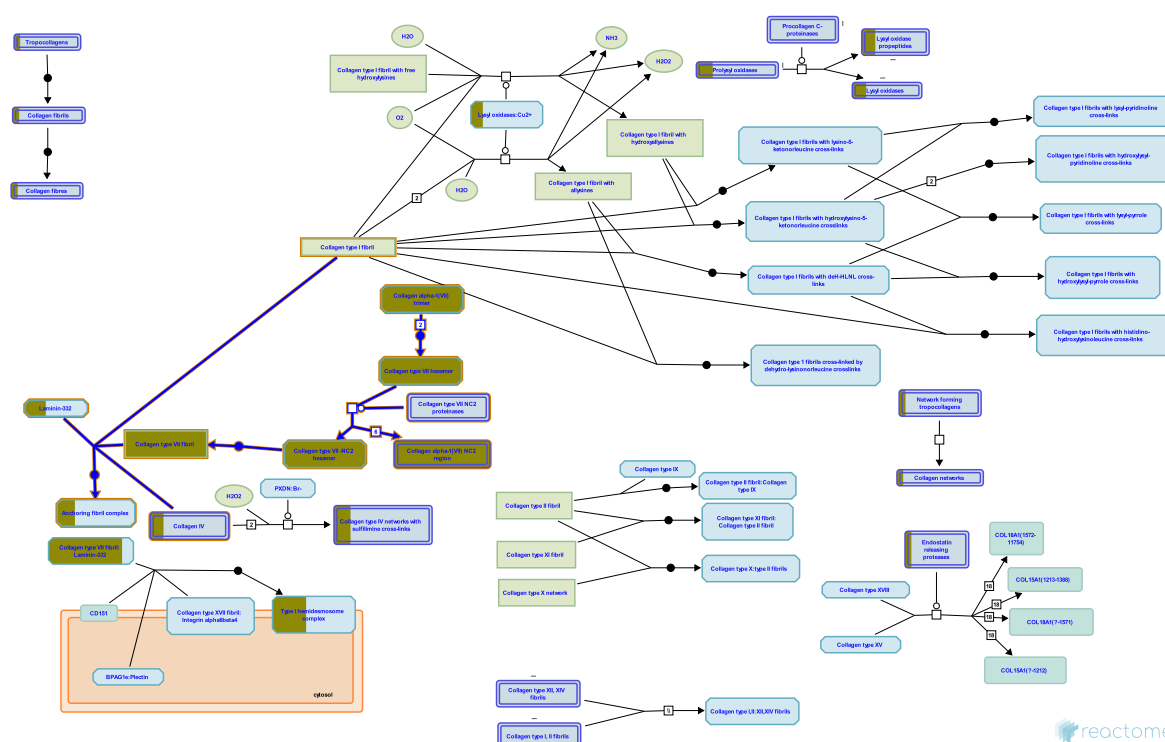
Edit history

Date	Action	Author
2017-07-24	Created	Orlic-Milacic M
2020-07-14	Authored	Rothfels K, Orlic-Milacic M
2021-02-05	Reviewed	Fort P
2021-02-25	Edited	Orlic-Milacic M
2025-11-15	Modified	Weiser JD

16 submitted entities found in this pathway, mapping to 16 Reactome entities

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
ARHGAP26	Q9UNA1	ARHGAP42	A6NI28	ARHGEF39	Q8N4T4
BAIAP2L1	Q9UHR4	CAV1	Q03135	CHN1	P15882
DOCK4	Q8N1I0	EPHA2	P29317	JAG1	P78504
KALRN	O60229	NHS	Q6T4R5	PIK3R1	P27986
SLC1A5	Q15758	TMEM133	A6NI28	VAV3	Q9UKW4
WASF3	Q9UPY6				

21. Anchoring fibril formation (R-HSA-2214320)



Cellular compartments: extracellular region.

Collagen VII forms anchoring fibrils, composed of antiparallel dimers that connect the dermis to the epidermis (Bruckner-Tuderman 2009, Has & Kern 2010). During fibrillogenesis, the nascent type VII procollagen molecules dimerize in an antiparallel manner. The C-propeptide is then removed by Bone morphogenetic protein 1 (Rattenholl et al. 2002) and the processed antiparallel dimers laterally aggregate (Villone et al. 2008, Gordon & Hahn 2010).

References

Chung HJ & Uitto J (2010). Type VII collagen: the anchoring fibril protein at fault in dystrophic epidermolysis bullosa. *Dermatol Clin*, 28, 93-105. [🔗](#)

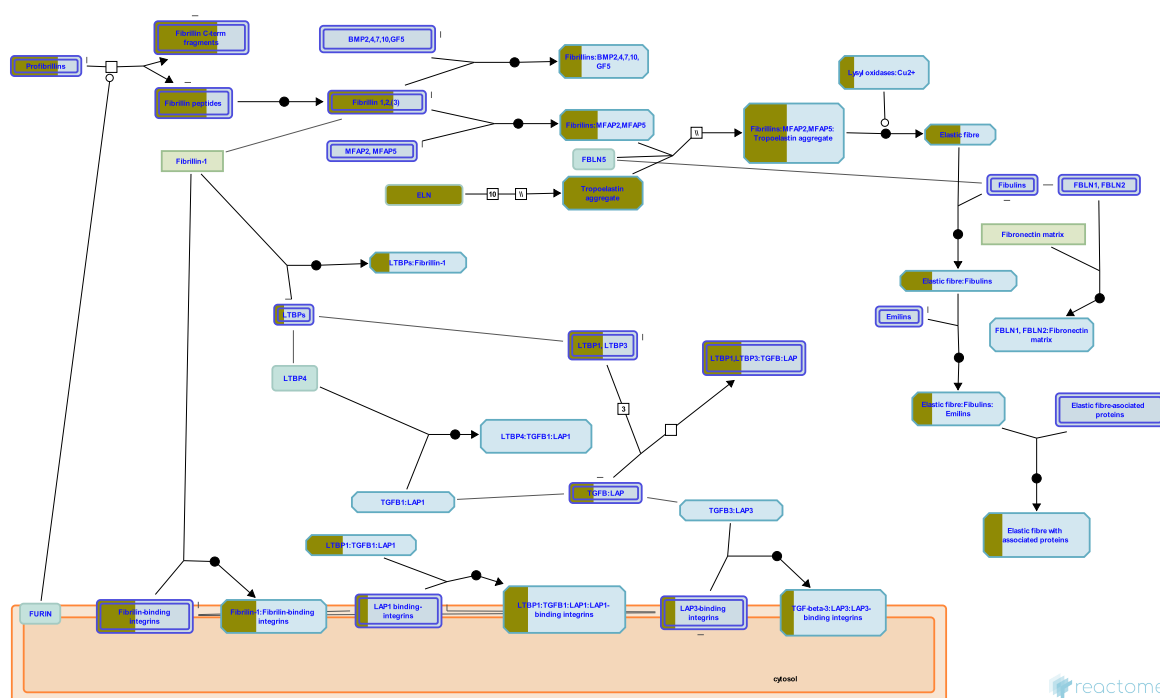
Edit history

Date	Action	Author
2012-04-30	Authored	Jupe S
2012-04-30	Created	Jupe S
2012-10-08	Reviewed	Kalamajski S, Raleigh S
2012-11-12	Edited	Jupe S
2012-11-19	Reviewed	Ricard-Blum S
2025-11-15	Modified	Weiser JD

2 submitted entities found in this pathway, mapping to 3 Reactome entities

Input	UniProt Id	Input	UniProt Id
COL7A1	P29400, Q02388	LAMA3	Q16787

22. Elastic fibre formation (R-HSA-1566948)



Cellular compartments: extracellular region.

Elastic fibres (EF) are a major structural constituent of dynamic connective tissues such as large arteries and lung parenchyma, where they provide essential properties of elastic recoil and resilience. EF are composed of a central cross-linked core of elastin, surrounded by a mesh of microfibrils, which are composed largely of fibrillin. In addition to elastin and fibrillin-1, over 30 ancillary proteins are involved in mediating important roles in elastic fibre assembly as well as interactions with the surrounding environment. These include fibulins, elastin microfibril interface located proteins (EMILINs), microfibril-associated glycoproteins (MAGPs) and Latent TGF-beta binding proteins (LTBPs). Fibulin-5 for example, is expressed by vascular smooth muscle cells and plays an essential role in the formation of elastic fibres through mediating interactions between elastin and fibrillin (Yanigasawa et al. 2002, Freeman et al. 2005). In addition, it plays a role in cell adhesion through integrin receptors and has been shown to influence smooth muscle cell proliferation (Yanigasawa et al. 2002, Nakamura et al. 2002). EMILINs are a family of homologous glycoproteins originally identified in extracts of aortas. Found at the elastin-fibrillin interface, early studies showed that antibodies to EMILIN can affect the process of elastic fibre formation (Bressan et al. 1993). EMILIN1 has been shown to bind elastin and fibulin-5 and appears to coordinate their common interaction (Zanetti et al. 2004). MAGPs are found to co-localize with microfibrils. MAGP-1, for example, binds strongly to an N-terminal sequence of fibrillin-1. Other proteins found associated with microfibrils include vitronectin (Dahlback et al. 1990).

Fibrillin is most familiar as a component of elastic fibres but microfibrils with no elastin are found in the ciliary zonules of the eye and invertebrate circulatory systems. The addition of elastin to microfibrils is a vertebrate adaptation to high pulsatile pressures in their closed circulatory systems (Faury et al. 2003). Elastin appears to have emerged after the divergence of jawless vertebrates from other vertebrates (Sage 1982).

Fibrillin-1 is the major structural component of microfibrils. Fibrillin-2 is expressed earlier in development than fibrillin-1 and may be important for elastic fiber formation (Zhang et al. 1994). Fibrillin-3 arose as a duplication of fibrillin-2 that did not occur in the rodent lineage. It was first isolated from human brain (Corson et al. 2004).

Fibrillin assembly is not as well defined as elastin assembly. The primary structure of fibrillin is dominated by calcium binding epidermal growth factor like repeats (Kielty et al. 2002). Fibrillin may form dimers or trimers before secretion. However, multimerisation predominantly occurs outside the cell. Formation of fibrils appears to require cell surface structures suggesting an involvement of cell surface receptors. Fibrillin is assembled pericellularly (i.e. on or close to the cell surface) into microfibrillar arrays that undergo time dependent maturation into microfibrils with beaded-string appearance. Transglutaminase forms gamma glutamyl epsilon lysine isopeptide bonds within or between peptide chains. Additionally, intermolecular disulfide bond formation between fibrillins is an important contributor to fibril maturation (Reinhardt et al. 2000).

Models of fibrillin-1 microfibril structure suggest that the N-terminal half of fibrillin-1 is asymmetrically exposed in outer filaments, while the C-terminal half is buried in the interior (Kuo et al. 2007). Fibrillinopathies include Marfan syndrome, familial ectopia lentis, familial thoracic aneurysm, all due to mutations in the fibrillin-1 gene FBN1, and congenital contractural arachnodactyly which is caused by mutation of FBN2 (Maslen & Glanville 1993, Davis & Summers 2012).

In vivo assembly of fibrillin requires the presence of extracellular fibronectin fibres (Sabatier et al. 2009). Fibrillins have Arg-Gly-Asp (RGD) sequences that interact with integrins (Pfaff et al. 1996, Sakamoto et al. 1996, Bax et al., 2003, Jovanovic et al. 2008) and heparin-binding domains that interact with a cell-surface heparan sulfate proteoglycan (Tiedemann et al. 2001) possibly a syndecan (Ritty et al. 2003). Fibrillins also have a major role in binding and sequestering growth factors such as TGF beta into the ECM (Neptune et al. 2003). Proteoglycans such as versican (Isogai et al. 2002), biglycan, and decorin (Reinboth et al. 2002) can interact with the microfibrils. They confer specific properties including hydration, impact absorption, molecular sieving, regulation of cellular activities, mediation of growth factor association, and release and transport within the extracellular matrix (Buczek-Thomas et al. 2002). In addition, glycosaminoglycans have been shown to interact with tropoelastin through its lysine side chains (Wu et al. 1999), regulating tropoelastin assembly (Tu & Weiss 2008).

Elastin is synthesized as a 70kDa monomer called tropoelastin, a highly hydrophobic protein composed largely of two types of domains that alternate along the polypeptide chain. Hydrophobic domains are rich in glycine, proline, alanine, leucine and valine. These amino acids occur in characteristic short (3-9 amino acids) tandem repeats, with a flexible and highly dynamic structure (Floquet et al. 2004). Unlike collagen, glycine in elastin is not rigorously positioned every 3 residues. However, glycine is distributed frequently throughout all hydrophobic domains of elastin, and displays a strong preference for inter-glycine spacing of 0-3 residues (Rauscher et al. 2006).

Elastic fibre formation involves the deposition of tropoelastin onto a template of fibrillin rich microfibrils. Recent results suggest that the first step of elastic fiber formation is the organization of small globules of elastin on the cell surface followed by globule aggregation into microfibrils (Kozel et al. 2006). An important contribution to the initial stages assembly is thought to be made by the intrinsic ability of the protein to direct its own polymeric organization in a process termed 'coacervation' (Bressan et al. 1986). This self-assembly process appears to be determined by interactions between hydrophobic domains (Bressan et al. 1986, Vrhovski et al. 1997, Bellingham et al. 2003, Cirulis & Keeley 2010) which result in alignment of the cross-linking domains, allowing the stabilization of elastin through the formation of cross-links generated through the oxidative deamination of lysine residues, catalyzed by members of the lysyl oxidase (LOX) family (Reiser et al. 1992, Mithieux & Weiss 2005). The first step in the cross-linking reaction is the oxidative formation of the delta aldehyde, known as alpha aminoadipic semialdehyde or allysine (Partridge 1963). Subsequent reactions that are probably spontaneous lead to the formation of cross-links through dehydrolysinonorleucine and allysine aldol, a trifunctional cross-link dehydromerodesmosine and two tetrafunctional cross-links desmosine and isodesmosine (Lucero & Kagan 2006), which are unique to elastin. These cross-links confer mechanical integrity and high durability. In addition to their role in self-assembly, hydrophobic domains provide elastin with its elastomeric properties, with initial studies suggesting that the elastomeric properties of elastin are driven through changes in entropic interactions with surrounding water molecules (Hoeve & Flory 1974).

A very specific set of proteases, broadly grouped under the name elastases, is responsible for elastin remodelling (Antonicelli et al. 2007). The matrix metalloproteinases (MMPs) are particularly important in elastin breakdown, with MMP2, 3, 9 and 12 explicitly shown to degrade elastin (Ra & Parks 2007). Nonetheless, elastin typically displays a low turnover rate under normal conditions over a lifetime (Davis 1993).

References

Kozel BA, Rongish BJ, Czirok A, Zach J, Little CD, Davis EC, ... Mecham RP (2006). Elastic fiber formation: a dynamic view of extracellular matrix assembly using timer reporters. *J Cell Physiol*, 207, 87-96. [🔗](#)

Edit history

Date	Action	Author
2011-09-06	Created	Jupe S
2012-04-30	Authored	Jupe S
2012-11-02	Reviewed	Muiznieks LD
2012-11-12	Edited	Jupe S
2013-02-27	Reviewed	Parkinson J
2025-11-15	Modified	Weiser JD

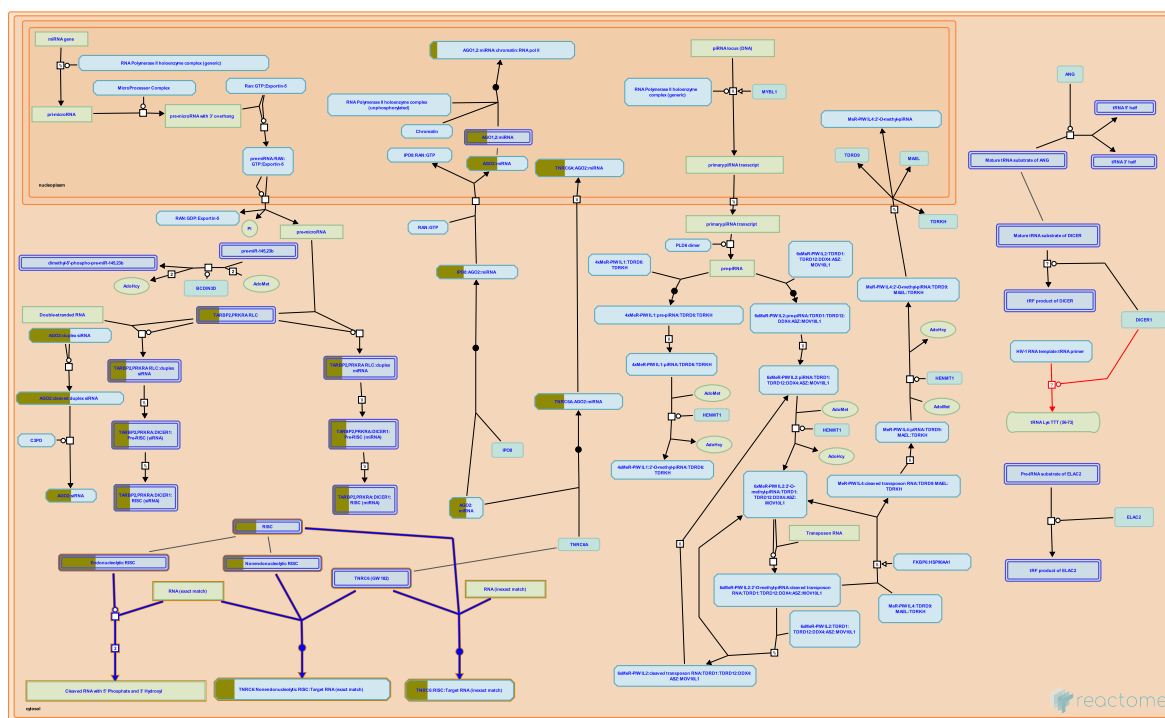
7 submitted entities found in this pathway, mapping to 8 Reactome entities

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ELN	P15502	FBN2	P35556, Q75N90	ITGA5	P08648
ITGB3	P05106	LOXL4	Q96JB6	LTBP1	Q14766
TGFB2	P61812				

Interactors found in this pathway (1)

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
ELN	P15502	Q9UBX5			

23. Post-transcriptional silencing by small RNAs (R-HSA-426496)



Cellular compartments: cytosol.

Small RNAs act with components of the RNA-induced silencing complex (RISC) to post-transcriptionally repress expression of mRNAs (reviewed in Nowotny and Yang 2009, Chua et al. 2009). Two mechanisms exist: 1) cleavage of target RNAs by complexes containing Argonaute2 (AGO2, EIF2C2) and a guide RNA that exactly matches the target mRNA and 2) inhibition of translation of target RNAs by complexes containing AGO2 and an inexact matching guide RNA or by complexes containing a nonendonucleolytic Argonaute (AGO1 (EIF2C1), AGO3 (EIF2C3), or AGO4 (EIF2C4)) and a guide RNA of exact or inexact match. Small interfering RNAs (siRNAs) and microRNAs (miRNAs) can serve as guide RNAs in both types of mechanism.

RNAi also appears to direct chromatin modifications that cause transcriptional gene silencing (reviewed in Verdel et al. 2009).

References

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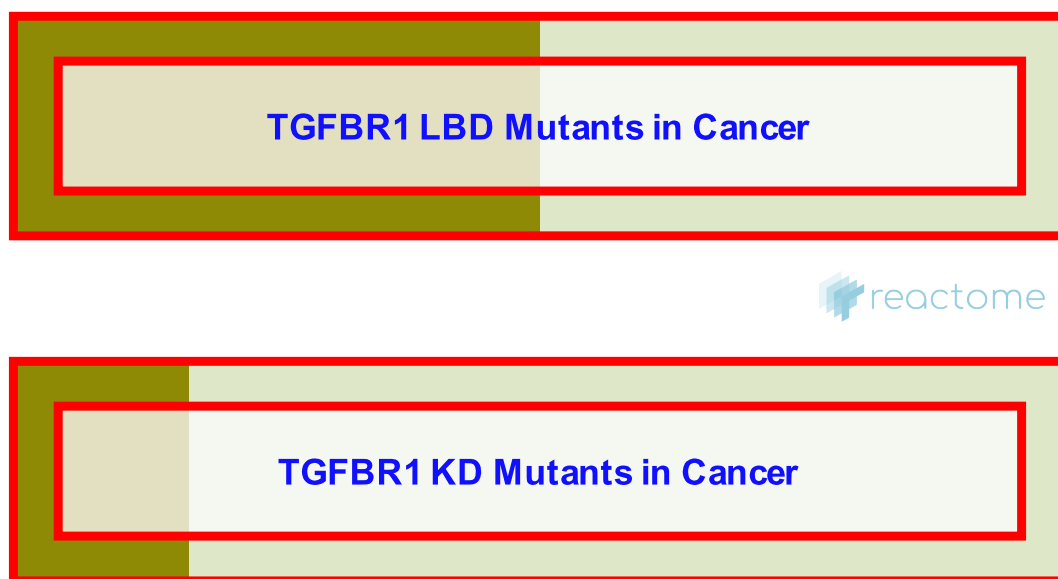
Edit history

Date	Action	Author
2009-06-10	Edited	May B
2009-06-10	Authored	May B
2009-06-17	Created	May B
2012-02-11	Reviewed	Tomari Y
2025-11-15	Modified	Weiser JD

1 submitted entities found in this pathway, mapping to 2 Reactome entities

Input	UniProt Id
AGO2	Q9H9G7, Q9UKV8

24. Loss of Function of TGFBR1 in Cancer ([R-HSA-3656534](#))



Diseases: cancer.

TGF-beta receptor 1 (TGFBR1) loss-of-function is a less frequent mechanism for inactivation of TGF-beta signaling in cancer compared to SMAD4 and TGFBR2 inactivation. Genomic deletion of TGFBR1 locus has been reported in pancreatic cancer (Goggins et al. 1998), biliary duct cancer (Goggins et al. 1998) and lymphoma (Schiemann et al. 1999), while loss-of-function mutations have been reported in breast (Chen et al. 1998) and ovarian cancer (Chen et al. 2001), metastatic head-and-neck cancer (Chen et al. 2001), and in Ferguson-Smith tumors (multiple self-healing squamous epithelioma - MSSE) (Goudie et al. 2011). Loss-of-function mutations mainly affect the ligand-binding extracellular domain of TGFBR1 and the kinase domain of TGFBR1 (Goudie et al. 2011). In the mouse model of colorectal cancer, *Tgfr1* haploinsufficiency cooperates with *Apc* haploinsufficiency in the development of intestinal tumors (Zeng et al. 2009).

References

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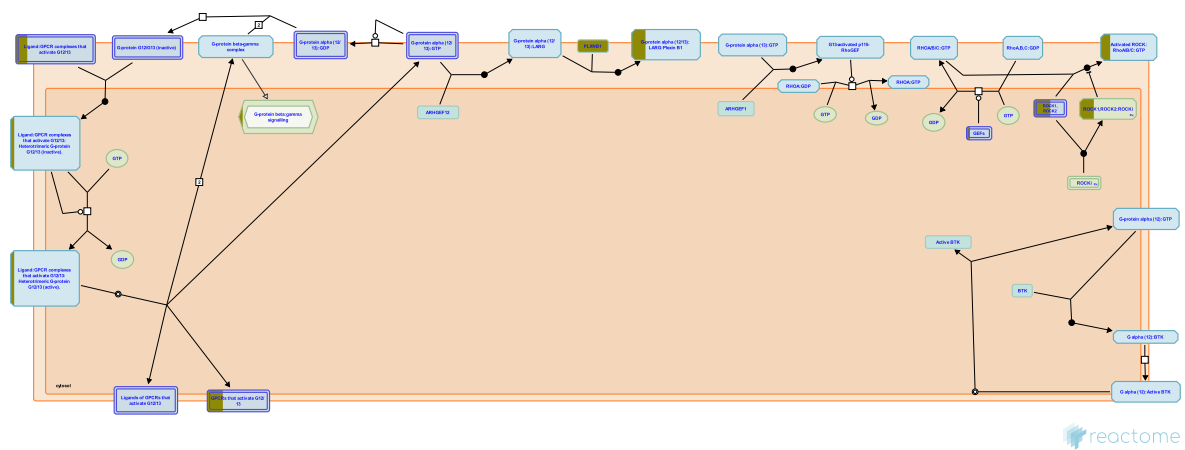
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Date	Action	Author
2013-06-05	Created	Orlic-Milacic M
2013-08-08	Edited	Orlic-Milacic M
2013-08-08	Reviewed	Meyer S, Akhurst RJ
2013-08-08	Authored	Meyer S, Akhurst RJ, Orlic-Milacic M
2023-10-12	Modified	Weiser JD

2 submitted entities found in this pathway, mapping to 2 Reactome entities

Input	UniProt Id	Input	UniProt Id
FPR1	P62942	TGFBR2	P37173

25. G alpha (12/13) signalling events (R-HSA-416482)



Cellular compartments: cytoplasmic side of plasma membrane.

The G12/13 family is probably the least well characterized subtype, partly because G12/13 coupling is difficult to determine when compared with the other subtypes which predominantly rely on assay technologies that measure intracellular calcium. The G12/13 family are best known for their involvement in the processes of cell proliferation and morphology, such as stress fiber and focal adhesion formation. Interactions with Rho guanine nucleotide exchange factors (RhoGEFs) are thought to mediate many of these processes. (Buhl et al.1995, Sugimoto et al. 2003). Activation of Rho or the regulation of events through Rho is often taken as evidence of G12/13 signaling. Receptors that are coupled with G12/13 invariably couple with one or more other G protein subtypes, usually Gq.

References

Worzfeld T, Wettschureck N & Offermanns S (2008). G(12)/G(13)-mediated signalling in mammalian physiology and disease. Trends Pharmacol Sci, 29, 582-9. [🔗](#)

Suzuki N, Hajicek N & Kozasa T (2009). Regulation and physiological functions of G12/13-mediated signaling pathways. Neurosignals, 17, 55-70. [🔗](#)

Edit history

Date	Action	Author
2009-03-18	Authored	Jupe S
2009-03-27	Edited	Jupe S
2009-03-27	Created	Jupe S
2009-06-03	Reviewed	Akkerman JW

10 submitted entities found in this pathway, mapping to 10 Reactome entities

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
ADRA1B	P35368	AKAP13	Q12802	ARHGEF37	A11GU5
ARHGEF39	Q8N4T4	FGD1	P98174	FGD4	Q96M96
KALRN	O60229	PLXNB1	O43157	ROCK1	Q13464
VAV3	Q9UKW4				

Interactors found in this pathway (1)

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
SH3BP5	O60239	Q06187			

6. Identifiers found

Below is a list of the input identifiers that have been found or mapped to an equivalent element in Reactome, classified by resource.

472 of the submitted entities were found, mapping to 629 Reactome entities

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
ABCA6	Q8N139	ABLIM2	Q6H8Q1	ABLIM3	O94929
ACACA	O00763, Q13085	ACADL	P28330	ACAN	P16112
ACKR1	Q16570	ACSL1	P33121, Q9ULC5	ACSM5	Q6NUN0
ACVR1C	Q8NER5	ACVR2A	P27037	ACVRL1	P37023
ADAP1	O75689	ADARB1	P78563	ADH1B	P00325, P28332
ADI1	Q9BV57	ADORA1	P30542	ADORA2B	P29275
ADRA1B	P35368	ADRA2A	P08913	AGO2	Q9H9G7, Q9UKV8
AGTR1	P30556	AHRR	A9YTQ3	AKAP13	Q12802
ALCAM	Q13740	ALOX15B	O15296	ALPL	P05186
AMOT	Q4VCS5-2	ANGPTL4	Q9BY76	ANKRD1	Q15327
ANKRD27	Q96NW4	ANO3	Q9BYT9	ANPEP	P15144
AOX1	Q06278	APLN	Q9ULZ1	APOB	P04114
APOD	P05090	APOOL	Q6UXV4	ARHGAP26	Q9UNA1
ARHGAP42	A6NI28	ARHGEF37	A1IGU5	ARHGEF39	Q8N4T4
ARL4C	P56559	ARMC8	Q8IUR7	ATP10D	Q9P241
ATP1A1	P05023	ATP6V1A	P38606	ATP8B1	O43520
B3GNT2	Q9NY97	B3GNT5	Q9BYG0	BAIAP2L1	Q9UHR4
BCL2L1	Q07817	BCL9	O00512, Q9Y3Y4	BDKRB1	P46663
BDKRB2	P30411, P46663	BIRC3	Q13489	BLOC1S6	Q9UL45
BTG2	P78543	BTN2A2	Q8WVV5	C5AR1	P21730
C5AR2	Q9P296	CACNB2	Q08289	CADM3	Q8N126
CASP3	P42574	CASP6	Q9BZM1	CAV1	Q03135
CBS	P35520	CCDC22	O60826	CCL7	P80098
CCND3	P30281	CD163	Q86VB7	CDC42EP3	Q9UKI2
CDH13	P55290	CDH9	Q9ULB4	CEBPB	P17676
CERS6	Q6ZMG9	CFLAR	O15519-1, O15519-2	CH25H	O95992
CHAC1	Q9BUX1	CHML	P26374	CHN1	P15882
CHST2	Q9Y4C5	CHST7	Q9NS84	CIDEC	Q96AQ7-1
CILP	O75339	CISH	Q9NSE2	CITED4	Q96RK1
CLCF1	Q9UBD9	CLTCL1	P53675	CNIH3	Q8TBE1
COL5A3	P25940	COL7A1	P29400, Q02388	COMMD7	Q86VX2
COMP	P49747	COPS8	Q99627	CORIN	Q9Y5Q5
CPM	P14384	CPNE7	Q9UBL6	CPPED1	Q9BRF8
CRISPLD2	Q9H0B8	CRLF1	O75462	CRYAB	P02511
CSAD	Q9Y600	CSF1	P09603	CXCL1	P09341
CXCL12	P48061	CXCL2	P19875	CXCL3	P19875
CXCL8	P10145	CYLD	Q9NQC7	CYP19A1	P11511

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
CYP24A1	Q07973	CYTH3	O43739	DAAM1	Q9Y4D1
DAB2	P98082	DCUN1D3	Q8IWE4	DENND3	A2RUS2
DENND5B	Q6ZUT9	DGKI	O75912	DHRS7B	Q6IAN0
DKK1	O94907	DKK2	Q9UBU2	DNAJC6	O75061
DNM3	Q9UQ16	DNMT3A	Q9Y6K1	DNMT3B	Q9UBC3
DOCK4	Q8N1I0	DSE	Q9UL01	DSP	P15924
DUSP1	P28562	DUSP10	Q9Y6W6	DUSP5	Q16690
DYRK2	Q92630	ECE1	P42892	EDNRB	P24530
EHHADH	Q08426	ELL2	O00472	ELN	P15502
ELOVL3	Q9HB03	EML4	Q9HC35	EPHA2	P29317
EPHA4	P54764	EPHB3	P54753	EPHB6	O15197
EPS8	Q12929	ERBB3	P21860-1	ERG	Q12809
ETS2	P15036	EXTL2	Q9UBQ6	F2RL1	P25116
FAM20A	Q96MK3	FAM20C	Q8IXL6	FBN2	P35556, Q75N90
FGD1	P98174	FGD4	Q96M96	FGF2	P09038
FGL2	Q14314	FITM2	A5D6W6, Q8N6M3	FKBP5	Q13451
FLG	P11362-1, P11362-19	FLRT2	O43155	FMO2	Q99518
FNBP1L	Q5T0N5	FOXC2	Q99958	FOXO1	Q12778
FOXO3	O43524	FOXP1	Q9H334-8	FPR1	P21462
FST	P19883	FSTL3	O95633	FZD6	O60353
GALNT15	Q7Z4T8, Q8N3T1	GAS1	P54826	GBAS	O75323
GCLC	P48506	GCNT1	Q02742	GDPD5	Q8WTR4
GEM	P55040	GJA1	P17302	GLI1	P08151
GLUL	P15104	GMPR	P36959	GPER1	Q99527
GPR116	Q8IZF2	GPR176	Q14439	GPR39	O43194
GPX3	O75715, P22352	GRIA1	P42261	GRIP2	Q9C0E4
GRK5	P34947	GSTM4	Q03013	HAS2	Q92819
HCCS	P53701	HDAC7	Q8WUI4	HIF3A	Q9Y2N7
HIP1	O00291	HIP1R	O75146	HIPK2	Q9H2X6
HMOX1	P09601	HOXA3	O43365	HOXB3	P14651
HS3ST5	Q8IZT8, Q96QI5	HSD11B1	P28845	HSD17B14	Q9BPX1
HTR1D	P28221	HTR2B	P41595	ICAM1	P05362
IFNGR1	P15260	IGF2BP2	Q9Y6M1	IGFBP2	P18065
IKBKE	Q14164	IL11	P20809	IL16	Q14005
IL17RC	Q8NAC3	IL1B	P01583, P01584	IL34	Q6ZMJ4
IL4I1	Q96RQ9	IL6	P05231	IL6ST	P40189
IMPA2	O14732	INHBB	P09529	INHBE	P58166
IRAK3	Q9Y616	IRS2	Q9Y4H2	ISM1	Q9NSE4
ITGA10	O75578	ITGA5	P08648	ITGB3	P05106
ITPK1	Q13572	ITPKC	Q96DU7	JAG1	P78504
KALRN	O60229	KCNB1	Q14721	KCNE4	Q8WWG9
KCNIP3	Q9Y2W7	KCNJ15	Q99712	KCNK6	Q9Y257
KCNS2	Q9ULS6	KDM2B	Q8NHM5	KIF13A	Q9H1H9
KLF15	Q9UIH9	KLF4	O43474	KLF9	Q13886

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
KRTAP1-5	Q07627, Q9BYS1	KRTAP2-3	P0C7H8	LAMA3	Q16787
LAMA4	Q16363	LEP	P41159	LGR4	Q9BXB1
LHFPL2	Q6ZUX7	LHX9	Q9NQ69	LIF	P15018
LIMA1	Q8TDZ2	LMCD1	Q9NZU5	LMO3	P25800
LOXL4	Q96JB6	LPCAT2	Q7L5N7	LPIN3	Q9BQK8
LRRC8A	Q8IWT6	LRRFIP1	Q32MZ4	LTBP1	Q14766
LYPD1	Q8N2G4	MAFF	Q9ULX9	MAP2	P50579
MAP2K6	P52564	MAP3K5	Q99683	MAPK13	O15264
MARCKS	P29966	MBOAT2	Q6ZWT7	MCL1	Q07820
METTL7A	Q9H8H3	MGLL	Q99685	MIB1	Q86YT6
MID1	Q9NQW6	MMP16	P51512	MMP7	P09237
MT1X	P80297	MTHFD2	P13995	MYC	P01106
MYH3	P11055	NCKIPSD	Q9NZQ3	NCOA3	Q9Y6Q9
NDRG1	Q92597	NEK7	Q8TDX7	NF1	P21359
NFIX	Q14938	NFKBIA	P25963	NFKBIE	O00221
NHS	Q6T4R5	NIPAL2	Q9H841	NLGN2	Q8NFZ4
NLGN4Y	Q8NFZ3	NOG	Q13253	NOL4L	Q86VX2
NPAS2	Q99743	NPL	Q9BXD5	NPR1	P16066
NPY2R	P49146	NR2F1	P10589	NR3C1	P04150-1, P04150-2, P04150-3, P04150-4, P04150-5, P04150-6, P04150-7, P04150-8, P04150-9
NRCAM	Q92823	NRG1	Q02297, Q02297-10	NTRK1	P04629
NUDT4	Q96G61, Q9NZJ9	OLAH	Q9NV23	OMD	Q99983
OPN3	Q9H1Y3	OPTN	Q96CV9	OSBPL8	Q9BZF1
OSR1	Q8TAX0	P2RY6	Q15077	PARP8	Q8N3A8
PCGF5	Q86SE9	PDE1C	Q01064, Q14123	PDGFC	Q9NRA1-1
PDGFD	Q9GZP0	PDK4	Q16654	PEAR1	P11047
PER1	O15534	PER3	P56645	PGRMC2	O15173
PHC2	P78364, Q8IXK0	PI4K2A	Q9BTU6	PIGV	Q9NUD9
PIK3R1	P27986	PITX2	Q99697	PLA2G4A	P47712
PLAU	P00749	PLCE1	Q9P212	PLXNA2	O75051
PLXNB1	O43157	PNPLA2	Q96AD5	POU2F2	P09086
PPFIA4	O75335	PPP1R14A	Q96A00	PRG4	Q96GM1
PRICKLE1	Q96MT3	PRKD1	Q15139	PTGER2	P35408, P43116
PTGIR	P43119	PTGS1	P23219	PTHLH	P12272
PTK2B	Q14289	PTPRB	P23467	PTPRK	Q15262
PTPRS	Q13332	PTX3	P26022	PYGB	P11216
RAB30	Q15771	RALA	P11233	RAMP1	O60894
RAP2B	P61225	RARG	P13631	RASA1	P20936
RASGRP2	Q7LDG7	RBFOX2	O43251	RDH11	Q96NR8
REV3L	O60673	RGCC	Q9H4X1	RGMB	Q6NW40
RGS3	P49796	RGS9	O75916	ROCK1	Q13464
ROR2	Q01974	RPS6KA3	P51812	RSPO1	Q2MKA7
RSPO3	Q9BXY4	RTKN	Q9BST9	RUNX1	Q01196
S1PR1	P21453	SAA1	P0DJ18	SALL4	Q9UJQ4
SAMHD1	Q9Y3Z3	SBSPON	Q8IVN8	SCN7A	Q01118
SCN8A	Q9UQD0	SCNN1A	P37088	SEMA3A	Q14563

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
SEMA5A	Q13591	SERINC5	Q86VE9	SERPINA3	P01011
SGK223	Q86YV5	SIGLEC15	Q6ZMC9	SIRPA	P78324, Q9P1W8
SIX1	Q15475	SLA	Q13239	SLC11A2	P49281
SLC14A1	Q13336	SLC1A5	Q15758	SLC25A20	O43772
SLC26A6	Q9BXS9	SLC28A3	Q9HAS3	SLC29A4	Q7RTT9
SLC2A12	Q8TD20	SLC44A1	Q8WWI5	SLC4A5	Q9BY07
SLC5A6	Q9Y289	SLC6A6	P31641	SLC7A11	Q9UPY5
SMARCD2	Q92925	SNAI1	O95863	SNAI2	O43623
SOC5	O75159	SORT1	Q99523	SOX13	Q9UN79
SOX9	P48436	SP110	P23497	SPINT2	O43291
SPRED3	Q2MJR0	SQRDL	Q9Y6N5	SSB	P05455
STC2	O76061	STMN2	Q93045	STX3	Q13277
SUFU	Q9UMX1	SUOX	P51687	SUPT20H	Q8NEM7
SV2A	Q7L0J3	SYNJ2	O15056	SYT1	P21579, Q8NBV8
SYTL2	Q9HCH5	TAF4B	Q92750	TEAD3	Q99594
TFAP2A	P05549	TFEB	P19484	TGFB2	P61812
TGFBR2	P37173	TGIF1	Q15583	THRB	P10828
THSD1	Q9NS62	TJP2	Q9UDY2	TLE1	Q04724
TLR2	O60603	TLR4	O00206	TMEM133	A6NI28
TNFAIP3	P21580	TNFAIP8L3	Q5GJ75	TNFRSF19	Q9HAV5
TNFSF15	O95150	TNFSF9	P41273	TNIP1	Q15025
TOP2B	Q02880	TP63	Q9H3D4-2	TRAF1	Q13077
TRIM2	Q9C040	TRIM35	Q9UPQ4	TRIM37	O94972
TRIM6	Q9C030	TRIP10	Q15642	TRPC4	Q9UBN4
TRPC6	Q9Y210	TSC22D3	Q99576	TUBA4A	P68366
TXNIP	Q9H3M7	UGCG	Q16739	UGP2	Q16851
UPP1	Q16831	UST	Q9Y2C2	VASH2	Q86V25
VAV3	Q9UKW4	VEGFC	P49767	WASF3	Q9UPY6
WDR26	Q9H7D7	WNT7B	P56706	XPNPEP2	O43895
ZBTB16	Q05516	ZEB1	P37275	ZFP36	P26651
ZFP36L2	P47974				

Input	Ensembl Id	Input	Ensembl Id	Input	Ensembl Id
ABCA6	ENSG00000154262	ACACA	ENSG00000278540	ACSL1	ENSG00000151726
ADAP1	ENSG00000105963	ANGPTL4	ENSG00000167772	ANKRD1	ENSG00000148677
ANPEP	ENST00000300060	APOD	ENSG00000189058	ARL4C	ENSG00000188042, ENST00000339728
ATP6V1A	ENSG00000114573	BCL2L1	ENSG00000171552	BTG2	ENSG00000159388
CASP6	ENSG00000138794	CAV1	ENSG00000105974	CD163	ENSG00000177575
CEBPB	ENSG00000172216	CHAC1	ENSG00000128965	CIDEC	ENSG00000187288
CISH	ENSG00000114737	CSF1	ENSG00000184371	CXCL1	ENSG00000163739
CXCL12	ENSG00000107562	CXCL2	ENSG00000081041	CXCL8	ENSG00000169429
DCUN1D3	ENST00000324344	DDK1	ENSG00000107984	ELL2	ENST00000237853
ELOVL3	ENSG00000119915	EPHA4	ENSG00000116106	ETS2	ENSG00000157557
EXTL2	ENSG00000162694	FGF2	ENSG00000138685	FKBP5	ENSG00000096060
FOXO1	ENSG00000150907	FOXO3	ENSG00000118689	FPR1	ENSG00000171051
GCLC	ENSG00000001084	GEM	ENSG00000164949	GLI1	ENSG00000111087
GMPR	ENSG00000137198	H19	ENST00000252900	HMOX1	ENSG00000100292

Input	Ensembl Id	Input	Ensembl Id	Input	Ensembl Id
HOXA3	ENSG00000105997, ENST00000317201	HOXB3	ENSG00000120093, ENST00000311626	HSPB2	ENSG00000170276
ICAM1	ENSG00000090339	IL1B	ENSG00000125538	IL6	ENSG00000136244
ITGA5	ENSG00000161638	JAG1	ENSG00000101384	KLF15	ENSG00000163884
KLF4	ENSG00000136826	LEP	ENSG00000174697	LIF	ENSG00000128342
MCL1	ENSG00000143384	MGLL	ENSG00000074416	MID1	ENSG00000101871
MYC	ENSG00000136997	NDRG1	ENSG00000104419	NPAS2	ENSG00000170485
OSR1	ENSG00000143867	PDK4	ENSG00000004799	PER1	ENSG00000179094
PER3	ENSG00000049246	PIK3R1	ENSG00000145675	PITX2	ENSG00000164093
PNPLA2	ENSG00000177666	RALA	ENSG00000006451	RGCC	ENSG00000102760
RSPO3	ENSG00000146374	RUNX1	ENSG00000159216, ENST00000344691	S1PR1	ENSG00000170989
SAA1	ENSG00000173432	SALL4	ENSG00000101115	SAMHD1	ENSG00000101347
SIX1	ENSG00000126778	SLC7A11	ENSG00000151012	SNAI1	ENSG00000124216
SNAI2	ENSG00000019549	SOX9	ENST00000245479	TFAP2A	ENST00000379613
TP63	ENST00000354600	TRIM2	ENSG00000109654	TRIM35	ENSG00000104228
TRIM6	ENSG00000121236	ZEB1	ENSG00000148516		

Input	miRBase Id
MIR214	MI0000290

Interactors (474)

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
ABLIM2	Q6H8Q1-8	P35240	ABLIM3	O94929-2	Q13573
ACACA	Q13085	Q13085	ACADL	P28330	P22735
ACAN	P16112	P05067	ACVR2A	P27037	P43115
ADAP1	O75689	Q08379	ADARB1	P78563-4	P19525
ADI1	Q9BV57	P50281	ADORA1	P30542	P07550, P08588
ADORA2B	P29275	P29274	ADRA1B	P35368	P61073
AGO2	Q9UKV8	P04156	AGTR1	P30556	Q03135
AHDC1	Q5TGY3	Q13185	AKAP13	Q12802	P31946
ALCAM	Q13740	P17931	ALOX15B	O15296	P19474
AMOT	Q4VCS5	Q8N3R9	ANGPTL4	Q9BY76	Q14790
ANKRD1	Q9CR42	P19338	ANKRD27	Q96NW4	P25963
APBB2	Q92870-2	Q6VY07	APLN	APLN	O15354
APOB	P04114	P01130	APOD	P05090	Q96A29
APOL3	O95236-2	Q16585	APOOL	Q6UXV4	P46059
ARHGAP26	Q9UNA1	A1A4S6	ARHGAP42	A6NI28	P46108
ARHGEF39	Q8N4T4	Q2T9J0	ARL4C	P56559	P52306
ARMC8	Q8IUR7	Q6VN20, Q96S59	ARMC9	Q7Z3E5	P21709
ARRDC1	Q8N5I2	Q96PU5	ARRDC2	Q8TBH0	Q9H0M0
ATP1A1	P05023	P12931	ATP6V1A	P38606	Q9UII2
BAIAP2L1	Q9UHR4	Q9UQB8	BCL2L1	Q07817	Q9C000
BCL9	O00512	P35222	BIRC3	Q13489	Q13546, Q9Y572
BLOC1S6	Q9UL45	Q16623	BTG2	P78543	P10275
BTN2A2	Q8WVV5	P21579	C10orf55	Q5SWW7	P35680
C15orf39	Q6ZRI6	Q13257	C1QL1	O75973	O15498
C5AR1	P21730	P18031	C5AR2	Q9P296	P09601

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
CACNB2	Q08289	O14744	CASP3	P42574	P10599
CASP6	P55212	P18848	CAV1	Q03135	P19174
CBS	P35520	P17612	CCDC107	Q8WV48	Q9H1C4
CCDC22	O60826	Q13614	CCDC68	Q9H2F9	Q9Y3P9
CCND3	P30281	P29373	CCNDBP1	O95273	P09429
CDC42EP3	Q9UKI2	Q5S007	CDCP1	Q9H5V8	P12931
CDH13	P55290	Q9NUQ2	CDK14	O94921	P38936
CDK17	Q00537	P42772	CEBPB	P17676	Q13485
CERS6	Q6ZMG9	Q9H8X2	CFLAR	O15519, O15519-1	Q14790
CHML	P26374	P51159	CHN1	P15882	P28482
CHST7	Q9NS84	P40855	CISH	Q9NSE2	P00533
CLCF1	Q9UBD9, Q9UBD9-PRO_0000015616	P26992, O75462	CLTCL1	P53675	Q96FJ0
CNIH3	Q8TBE1	O15554	COL7A1	Q02388	Q5JRA6
COMP	P49747	O00244	COPS8	Q99627	Q15831
CORIN	Q9Y5Q5	O00501	CPNE7	Q9UBL6-2	Q9NRD5
CRLF1	O75462-PRO_0000011039, O75462	P26992, Q9UBD9	CRYAB	P02511	P12830
CSAD	Q9Y600	Q9NRF2-2	CSF1	P09603	P09603, P07333
CXCL1	P09341	Q13490	CXCL12	P48061	P78556, P13501, O15444, Q99616
CXCL2	P19875	P13501, P48061	CXCL8	P10145	P25025
CYLD	Q9NQC7	Q9Y6K9	CYTH3	O43739	Q13227
DAAM1	Q9Y4D1	P61586	DAB2	P98078, P98082	Q9UM54
DDIT4L	Q96D03	Q13158	DHRS7B	Q6IAN0	P13569
DIP2B	Q9P265	Q8TDQ0	DIXDC1	Q155Q3	O14641
DKK1	O94907	O75197	DNAJB4	Q9UDY4	Q9Y266
DNAJC6	O75061	P10636-8	DNM3	Q9UQ16	Q05193
DNMT3A	Q9Y6K1	P48552	DNMT3B	Q9UBC3	P48552
DOCK4	Q8N1I0	P62993	DRAM1	Q8N682	Q9NPL8
DSE	Q9UL01	Q9NRD5	DSP	P15924	P07332
DUSP1	P28562	P28482, P27361	DUSP10	Q9Y6W6	P60709
DUSP14	O95147	Q9BYZ6	DUSP23	Q9BVJ7	Q8TBC4
DUSP5	Q16690	P28482	DYRK2	Q92630	P23497
EBF3	Q9H4W6-2	Q9HAK2	ECE1	P42892	P17931
EHHADH	Q08426	Q16585	ELL2	O00472	P07550
ELN	P15502	Q9UBX5	EML2	A0A0C4DGQ7	P40937
EML4	Q9HC35	Q96PC5	EMP1	P54849	P25025
ENC1	O14682	Q13501	EPHA2	P29317	Q9UPY5
EPHA4	P54764	P05177	EPHB6	O15197-2	Q96RU7
EPS8	Q12929	P22681	ERBB3	P21860	Q14451
ERG	P11308	O00712, Q12857	ERGIC1	Q969X5	P04601
ERRFI1	Q9UJM3	P31749	ETS2	P15036	P17612
EVA1C	P58658	Q6PJF5	EXTL2	Q9UBQ6	P0DTC5
F2RL1	P55085	Q9BVG9	FAIM	Q9NVQ4	O43597
FAM107A	O95990-4, O95990	Q12933	FAM167A	Q96KS9	Q9UKG1
FAM168A	Q92567	O75177	FAM171B	Q6P995	P21802
FAM20C	Q8IXL6	P10451	FAM222B	Q8WU58	Q9UBE8

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
FAM43A	Q8N2R8	Q9BUR4	FAM84B	Q96KN1	O60551
FANK1	Q8TC84	P04792	FBN2	P35556	Q04721
FGD4	Q96M96	P17252	FGF2	P09038	O15530
FGL2	Q14314	Q9BYC5	FITM2	Q8N6M3	P22732
FKBP5	Q13451	Q9BQI3	FLG	P11362	P08908
FNBP1L	Q5T0N5	P78325	FOXC2	Q99958	Q99958
FOXN2	P32314	Q9Y297	FOXO1	Q12778	Q13485
FOXO3	O43524	Q13485	FOXP1	Q9H334	Q9BZS1
FPR1	P21462	O60427	FRAS1	Q86XX4	P08311
FRMD5	Q7Z6J6	Q92823	FRMD6	Q96NE9, Q96NE9-2	P40692
FST	P19883	P10599	GADD45B	O75293	O14733
GALNT15	Q8N3T1	O15552	GBAS	O75323	Q05513
GCLC	P48506	P48507	GCNT1	Q02742	Q07954
GDPD5	Q8WTR4	P43304	GEM	P55040	Q9BUH8
GJA1	P17302, P08050	Q07157	GLI1	P08151	Q9UMX1, Q96J02
GLUL	P15104	Q9BZE9	GPR116	Q8IZF2	Q9NPB3
GPR39	O43194	P08908	GPRASP2	Q96D09	P08311
GRIA1	P42261	Q9UIH9	GRIP2	Q9C0E4	Q15223
GRK5	P34947	P25963	GSTM4	Q03013	P05067
GTF2IRD1	Q9UHL9	Q6VMQ6	HCCS	P53701	P06307
HDAC7	Q9UKV0	Q06413	HEG1	Q9ULI3	P17931
HIF3A	Q9Y2N7-4	Q16665	HIP1	O00291	P35222
HIP1R	O75146	Q14627	HIPK2	Q9H2X6	Q09472
HMOX1	P09601	P0DTC3	HSD11B1	P28845	Q16880
HSD17B14	Q9BPX1	P29372	HSPB2	Q16082	Q9NQ94
HSPB7	Q9UBY9	Q9GZV8	HTRA3	P83110-2, P83110-1, P83110	P98170
ICAM1	P05362	P18440	IFNGR1	P15260	P23458
IGF2BP2	Q9Y6M1	Q9NZI8	IGFBP2	P18065	P59665
IKBKE	Q14164	P0DTC5	IL16	Q14005-2	Q14012
IL34	Q6ZMJ4-1	P07333	IL4I1	Q96RQ9	Q96CV9
IL6	P05231	P40189	IL6ST	P40189-1	P23458
IMPA2	O14732	Q8TBC4	INHBE	P58166	Q9Y2C2
INSC	Q1MX18	P55040	IRAK3	Q9Y616	Q9Y4K3
IRS2	Q9Y4H2	P27986	ITGA5	A0A061IJ01	P78423
ITGB3	P05106	P78423	ITPKC	Q96DU7	O14965
JAG1	P78504	P46531	JAGN1	Q8N5M9	P09601
KCNIP3	Q9Y2W7	P40189	KCNJ15	Q99712	P63252
KCNS2	Q9ULS6	P0DPK4	KDM2B	Q8NHM5	P63208
KIAA1522	Q9P206	Q9UQB8	KLF15	Q9UIH9	P12931
KLF4	O43474	O15550	KLF6	Q99612	Q04206
KRTAP1-5	Q9BYS1	P06850	KRTAP2-3	P0C7H8	Q8WUK0
LACTB	P83111	Q9NQ92	LAMA3	Q16787	P59665
LAMA4	Q16363	P04637	LANCL2	Q9NS86	Q9Y2W6
LDB2	O43679	Q6DKK2	LHFPL2	Q6ZUX7	Q14162
LHX9	Q9NQ69	P68400	LIF	P15018	P40189
LIMA1	Q9UHB6	P60953	LMO3	Q8TAP4-4	O94844
LMO4	P61968	Q13485	LOXL4	Q96JB6	P55212
LPCAT2	Q7L5N7	Q7Z5P4	LPHN2	O95490	P21926

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
LPIN3	Q9BQK8	Q14693	LRRRC8A	Q8IWT6	O43914
LRRFIP1	Q32MZ4	Q96LB3	LTBP1	Q14766	P09238
LYPD1	Q8N2G4	P41221	MAFF	Q9ULX9	O14867
MAP1A	P78559	Q9GZQ8	MAP2	P11137-4	P02649
MAP2K6	P52564	Q16539	MAP3K5	Q99683	P10599
MAP3K7CL	P57077	Q13227	MAPK13	O15264	P21462
MARCKS	P29966	P10636-8	MCL1	Q07820	Q13794
MEDAG	Q5VYS4	Q9H0E2	MEIS2	O14770-4	Q13882
METTL7A	Q9H8H3	P17252	MGLL	Q99685	P07550
MIB1	Q86YT6	Q9UHD2	MID1	P29372	P04156
MKX	Q8IYA7	Q9BZK7	MLLT11	Q13015	P04271
MMP7	P09237	P17931	MTHFD2	P13995	P49137
MYBPH	Q13203	Q13464	MYC	EBI-1265559	P40763
MYCL	P12524-2	P54646	MYH3	P11055	Q15475
NAV2	Q8IVL1	O75928	NCKIPSD	Q9NZQ3	Q9UQB8
NCOA3	Q9Y6Q9	Q09472	NDRG1	Q92597	Q9BYZ2
NEDD9	Q14511	Q8IZD9	NEK7	Q8TDX7	P00558
NEXN	Q0ZGT2	P60709	NF1	P21359	P21709
NFATC4	Q14934	P0CG48	NFIL3	Q16649	P18846
NFIX	Q14938-4, Q14938-3	P40763	NFKBIA	P25963	P61586
NFKBIE	O00221	P19838	NHS	Q6T4R5	O43639
NKD1	Q969G9	P67775	NLGN2	Q8NFX4	Q8N0Z6
NOL4L	Q96MY1	Q13363	NPAS2	Q99743	O00327
NPY2R	P49146	Q92538	NR2F1	P10589	Q00403
NR3C1	P06536	Q15788	NRCAM	Q92823	Q12860
NRG1	Q02297-10	P21854	NSMF	Q9EPI6	P27361
NTRK1	P04629	P04629	NUDT4	Q9NZJ9	P04792
NUDT6	P53370	P46199	NXPH3	O95157	Q9UMX1
NYNRIN	Q9P2P1	Q15185	OAF	Q86UD1	P56559
OLFML2A	Q68BL7	Q15848	OPTN	Q96CV9	Q96QE2
OSBPL8	Q9BZF1	O15530	OSR1	O95747	P02675
OSR2	Q8N2R0	P25788	P2RY6	Q15077	Q14974
PARP8	Q8N3A8	P61981	PCDHB13	Q9Y5F0	Q9UMR7
PCGF5	Q86SE9	P68400, P67870	PDGFC	Q9NRA1	P16234
PDGFD	Q9GZP0	O95714	PDK4	Q16654	Q15119
PDZRN3	Q9UPQ7	P31040	PER1	J3QSH9, O15534	P56645, Q16526, O15055
PER3	P56645	O15534	PGRMC2	O15173	Q6ZPD8
PHC2	Q8IXK0	P08047	PHLDB2	Q86SQ0	P21333
PI4K2A	Q9BTU6	Q96J02	PIGV	Q9NUD9	P37173
PIK3R1	P27986-2	P40763	PITX2	Q99697-3	Q12948
PKIA	P61925	P17612	PKIB	Q9C010-2	P17612
PLAU	P00749	P05121	PLEKHA7	Q6IQ23-2	Q15669
PLXNA2	O75051	Q01523, P59665	PLXNB1	O43157	Q92730
PNPLA2	Q96AD5	P09917	PNRC1	Q12796	Q9NPI6
POU2F2	P09086-4	P35548	PPP1R14A	Q96A00	P48729
PRG4	Q96GM1	Q9Y6B6	PRICKLE1	Q96MT3	Q9HAQ2
PRKD1	Q15139	P12830	PROSER2	Q86WR7	Q9Y316
PRRT2	Q7Z6L0	Q6UWP7	PSD3	Q9NYI0	P63104

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
PTGIR	P43119	Q7L8W6	PTGS1	P23219	P48645
PTK2B	Q14289	P40763	PTPDC1	A2A3K4	P46109
PTPRB	P23467	P27361	PTPRH	Q9HD43	P10912
PTPRK	Q15262	P17931	PTPRS	Q13332	Q16288
PTX3	P26022	P09038	PYGB	P11216	Q13451
RAB11FIP1	Q6WKZ4	P31946	RAB30	Q15771	Q9HD47
RALA	P11233	P13569	RALGPS2	Q86X27	P31947
RANBP3L	Q86VV4	Q08379	RAP2B	P61225	P19320
RARG	P13631	Q08999	RASA1	P20936, P50904	Q13480
RASL11A	Q6T310	P42858	RBFOX2	O43251	Q9NWB1
RCSD1	Q6JBY9	P52907	REV3L	O60673	Q9UI95
RGS3	P49796	P43155	RNF122	Q9H9V4	O95674
RNF24	Q9Y225-2	Q8WTS1	ROCK1	Q13464	P61586
ROR2	Q01974	O14503	RPS6KA3	P51812	P11362
RTKN	Q9BST9, Q8C6B2	P61586	RTKN2	Q8IZC4	Q96B97
RUNX1	Q01196	Q9H2X6	RUSC2	Q8N2Y8	P00533
S1PR1	P21453	Q7Z2K8	SALL2	Q9Y467	P08138
SAMHD1	Q9Y3Z3	P20248	SCNN1A	P37088	P46934
SDPR	O95810	Q03135	SEC14L5	O43304	P62333
SELENBP1	Q13228	Q9UJ68	SERPINA3	P01011	Q9UNE7
SERTAD2	Q14140	O60733	SGK223	Q86YV5	P46109
SH2D4A	Q9H788, Q9H788-2	O14936, Q9UJU6	SH3BP5	O60239	Q06187
SH3RF3	Q8TEJ3	Q13177	SIM1	P81133	Q9HBZ2
SIRPA	P78324	Q08722	SIX1	Q15475	Q99502
SLA	Q13239-3	Q13882	SLC11A2	P49281	P23634
SLC12A8	A0AV02	P26367	SLC14A1	Q13336-2	Q8WVV5
SLC1A5	Q15758	P0DTC5	SLC26A6	Q9BXS9-3	O43561-2
SLC2A12	Q8TD20	Q7Z7B1	SLC44A1	Q8WWI5	O14672
SLC7A11	Q9UPY5	P16070	SMARCD2	Q92925	O96019, Q15532
SMOC1	Q9H4F8	Q04721	SNAI1	O95863	Q09472
SNAI2	O43623	P21673	SNX30	Q8WV41	P48023
SOCS5	O75159	Q93034, Q9UBF6	SORT1	Q99523	P04629
SOX12	O60248	O00712, Q12857	SOX13	Q9UN79	Q12933
SOX9	P48436	Q12857	SPINT2	O43291	Q99640
SPRYD3	Q8NCJ5	P22692	SQRDL	Q9Y6N5	P01106
SSB	P05455	Q8WZA2	SSBP2	P81877	Q9UBH0
SSH2	Q76I76	P31946	STC2	O76061	P0DTC8
STMN2	Q93045	P40222	STX3	Q13277	P51809
SUFU	Q9UMX1-2, Q9UMX1	P08151	SUOX	P51687	Q15669
SUPT20H	Q8NEM7	Q16539	SV2A	Q02563	P0DPI0
SYNJ2	O15056	Q96B97	SYNPO	Q8N3V7	P04156
SYT1	P21579	P98155	SYTL2	Q9HCH5	P51159
TBC1D8	O95759	Q13077	TBC1D9	Q6ZT07	Q9HD23
TCEAL4	Q96EI5	Q93009	TCEANC2	Q96MN5	O94888
TEAD3	Q99594	P46937	TFAP2A	P05549	Q09472
TFEB	P19484	P31946	TGFB2	P61812	P05067
TGFBR2	P37173	P23458	TGIF1	Q15583	O15169

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
THRB	P10828	P60709	TJP2	Q9UDY2	Q07157
TLE1	Q04724	Q9H2X6	TLR2	O60603	Q99836
TLR4	O00206	P09429	TMEFF2	Q9UIK5	Q9Y5Z9
TMEM100	Q9NV29	P21964	TMEM133	A6NI28	P46108
TMEM164	Q5U3C3	P25103	TMTC1	Q8IUR5-4	P00488
TNFAIP3	P21580	Q14790	TNFAIP8L3	Q5GJ75	O00560
TNFRSF19	Q9NS68	P52565	TNFSF15	O95150-1	O95407
TNFSF9	P41273	Q14627	TNIP1	Q15025	P24387
TOP2B	Q02880	P46531	TP63	Q9H3D4	Q96J02
TRAF1	Q13077	P42772	TRERF1	Q96PN7	P46934
TRIM35	Q9UPQ4-2	Q9NPB3	TRIM37	O94972	O15169
TRIP10	Q15642	P48023	TSC22D3	Q99576-3, Q99576	Q9Y6D9
TUBA4A	P68366	P30622	TXNIP	Q9H3M7	P10599
UGCG	Q16739	P43119	UGP2	Q16851	O76003
USP53	Q70EK8	P46109	UST	Q9Y2C2	Q8WVC6
VASH2	Q86V25	P61981	VAV3	Q9UKW4	P62993
VEGFC	P97953-1	O60462	VSTM4	Q8IW00	Q16623
WASF3	Q9UPY6	Q13485	WDR26	Q9H7D7	O15169
WNT7B	P56706	P56705	ZAK	Q9NYL2	O75582
ZBTB16	Q05516	P05231	ZBTB18	Q99592	Q8WY64
ZEB1	P37275	Q92769	ZFP36	P26651	P16284
ZFP36L2	P47974	P31947	ZHX3	Q9H4I2, Q9H4I2-2	Q9UBE8
ZMAT4	Q9H898-2	Q15633	ZNF474	Q6S9Z5	P40692
ZXDC	Q2QGD7	P33076			
Input	ChEBI Id	Interacts with	Input	ChEBI Id	Interacts with
ANGPTL4	Q9BY76	28815	IL11	P20809	29108
SYT1	P21707	29036			

7. Identifiers not found

These 104 identifiers were not found neither mapped to any entity in Reactome.

ADPRH	AFAP1L1	APOL4	ARMC2	BAIAP2-AS1	BICC1	C11orf52	C15orf52
C1QTNF6	C2CD2	C3orf80	CADM3-AS1	CAMK2N1	CCDC69	CGREF1	CLMN
CPT1C	DBNDD2	DCHS1	DEPDC7	DIRC3	DKFZP586I1420	DNM3OS	EFTUD1
EPG5	FAM102B	FAM46C	FLG-AS1	FRMD4B	FRZB	GK5	GPR64
GPR88	GSAP	HPS3	HPS5	HSPB2-C11orf52	IFRD1	IGFLR1	KCTD2
KIAA0922	KLHL29	LDLRAD3	LINC00094	LINC00525	LINC00968	LINC01119	LINC01268
LOC100130539	LOC100132891	LOC100507387	LOC101927051	LOC643201	LYRM1	MAMSTR	MCTP1
METTL5	MEX3A	MIAT	MIR31HG	MOSPD1	MSC	Mar-10	NDNF
NEXN-AS1	NPR3	NT5DC3	NUDT4P1	NUDT4P2	PART1	PHTF1	PIEZO2
PPP1R14C	PRKY	PROSC	RASL11B	RASSF9	RERG	ROM1	SH3BP5-AS1
SH3RF3-AS1	SLC16A12	SLC16A4	SLC18B1	SLC38A6	SLC45A1	SLC46A3	SRRM3
SYPL2	TIMP4	TLDC2	TMEM135	TMEM138	TMEM178B	TMEM26	TOX
TPRA1	TRNP1	TTC13	TTPAL	TUBA3FP	ZNF365	ZNF804A	gene